

LaBinius – Fast Lattice-Based Binary Polynomial Commitment Scheme

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Abstract. Lattice-based proof systems are one of the leading routes to quantum-safe succinct proofs. Existing lattice-based schemes come with inherent limitations arising from the coupling between the commitment and constraint domains. First, commitment efficiency is restricted by the need for invertible challenge differences, which precludes the use of fully splitting moduli that enable the fastest lattice arithmetic. Second, the characteristic of the constraint field must equal the commitment modulus. This limits the efficiency of proving binary computations, including standard hash function evaluations.

We present **LaBinius**, a lattice-based polynomial commitment scheme, in which the evaluation arithmetic is separated from that of the commitment. The commitment runs over a composite modulus, and all of its prime factors are NTT-friendly. The evaluation claim lives in a binary extension field, without bitness checks or overhead over a binary arithmetisation.

We implement the scheme with AVX-512 kernels and connect it to the **Binius** and **Flock** front ends in an almost black-box way, which lets us use their arithmetisations tailored to binary fields. As a result, we obtain a SNARK for standard hashes. Proving Keccak-256, SHA-256 and BLAKE3 on one core, our prover takes at most 1.2 times as long as the hash-based prover supplied with each system, and we obtain a single-core throughput of over 67,000 BLAKE3 compressions per second with verification in under 40 milliseconds.

With **LaBRADOR** as a recursive compressor, the prover takes less than twice as long as the hash-based provers at the largest sizes we consider, while the proof size is below 100 KiB. This is about five times smaller than the proofs of **Binius** and **Flock** at those sizes and constitutes the smallest quantum-safe proof of a standard hash evaluation to date.

1 Introduction

Lattice-based proof systems have emerged as a promising class of practical quantum-safe succinct non-interactive arguments of knowledge (SNARKs). Compared with other post-quantum alternatives, lattice proofs offer leading proof sizes below 100 kilobytes [BS23, KK26, NOZ26, DBK⁺26]. Verifier runtimes can scale polylogarithmically and reach the single-digit millisecond range in practice [KLN⁺26]. Furthermore, prover runtimes remain practical even for very large instances.

A central metric for practical deployment is the total prover runtime, including both commitment and opening. This is particularly relevant in settings where data are continuously produced and proofs must be generated efficiently. In blockchain systems, for example, proof generation must keep pace with incoming transactions, so prover runtime can directly constrain throughput.

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Strong performance on selected algebraic statements, such as polynomial evaluations over prime fields, does not necessarily translate into efficient proofs of application-level computations, such as hash function evaluations. This distinction is particularly important to state-of-the-art lattice-based proof systems, including RoKoko [KLN⁺26], Akita [DBK⁺26], and Greyhound [NS24]. These systems support only arithmetisations over fixed finite fields whose characteristics have bit lengths typically between 30 and 50. Such fields are poorly suited to the bitwise operations prevalent in standard hash functions, which are more naturally expressed over $\mathbb{F}_2$ [DP25]. Akita provides a route to proving such computations through its integration with the Jolt zkVM [DBK⁺26, GTT26]. In this setting, however, the total prover runtime also includes the overhead of representing and executing the computation within the zkVM. Meaningful comparisons must therefore account for the full cost of proving the target computation, including arithmetisation, commitment, and opening.

The goal of this work is to substantially reduce the concrete prover runtimes of lattice-based proof systems while enabling flexible arithmetisations that are optimal for real-world tasks, without incurring any overhead.

Overview of lattice proofs. State-of-the-art lattice-based SNARK prover back-ends, e.g. RoKoko [KLN⁺26], can be divided into three main phases, similarly to non-lattice constructions. First, the prover commits to the witness using an algebraically structured commitment scheme. Then, in the sumcheck phase (typically in the literature captured abstractly as PIOP), the relations in the arithmetised proof statement are aggregated and reduced to a random evaluation of the witness when viewed as a large multilinear polynomial. More specifically, the polynomial interpolates the mr ring elements of the witness on the hypercube points $\{0, 1\}^{\log(mr)}$. Finally, the random evaluation is proven, which entails an opening proof for the commitment.

A lattice commitment is a simple high-dimensional linear map over a cyclotomic ring $\mathcal{R}_q$ of degree φ . The witness, consisting of mr ring elements with $mr\varphi$ integer coefficients in total, is formatted as a matrix $\mathbf{W} \in \mathcal{R}^{m \times r}$ with column vectors $\mathbf{w}_i \in \mathcal{R}^m$. Then, the commitment is given by $\mathbf{Y} = \mathbf{F}\mathbf{W}$ with subcommitments $\mathbf{y}_i = \mathbf{F}\mathbf{w}_i$, where $\mathbf{F}$ is a (fixed) public uniformly random matrix over $\mathcal{R}_q$. The commitment is binding under the SIS assumption if the ℓ_2 -norm of the witness is sufficiently small. Such a relation instead of being inspected directly may (and should) be recursed for better compression and verifier efficiency.

The sumcheck phase reduces low-degree relations on the witness to a polynomial evaluation. For compatibility with the lattice commitments the relations do not operate on the individual integer coefficients but instead on the ring elements inside the $\mathbf{w}_i$. In the opening and evaluation phase, the prover sends a random linear combination $\mathbf{v} = \mathbf{W}\mathbf{c} = c_0\mathbf{w}_0 + \dots + c_{r-1}\mathbf{w}_{r-1}$ of the $\mathbf{w}_i$ which is called the amortised opening. Succinctness stems from the fact that $\mathbf{v}$ is essentially r times smaller than the witness. The verifier checks that $\mathbf{v}$ is an opening for the amortised commitment, i.e. $\mathbf{F}\mathbf{v} \stackrel{?}{=} \mathbf{Y}\mathbf{c}$ and $\mathbf{v}$ is short.

Algebraic setup. In concrete instantiations of the commitment in the literature, the cyclotomic ring is chosen to be a power-of-two cyclotomic $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^\varphi + 1)$ modulo a rational prime q . More precisely, the degree φ is a small power of two in the range between 64 and 1024 and the size of the modulus q is in the range of 30 to 50 bits. A minimal modulus size of about 20 bits is needed for SIS-hardness, and slightly bigger sizes are chosen so that a negligible soundness error can be achieved more easily. Depending on the splitting behaviour of q , i.e. the factorisation of $x^\varphi + 1$ modulo q , $\mathcal{R}_q$ is isomorphic under the Chinese Remainder Theorem to a direct product of φ/l copies of the finite field $\mathbb{F}_{q^l}$. That is, $\mathcal{R}_q \cong \prod_{i=1}^{\varphi/l} \mathbb{F}_{q^l}$. This isomorphism is efficiently computed via the Number

Theoretic Transform (NTT) and used for implementing the ring arithmetic. The algebraic setup of the arithmetisation front-end and sumcheck back-end phase is inherited from that of the commitment. Hence, arithmetisation and sumcheck happen over finite fields of medium characteristic in the 30 to 50 bit range.

Concrete efficiency. A significant fraction of the prover runtime is spent on the commitment phase. Best possible commitment performance is achieved when the modulus q is fully-splitting meaning that the inertia degree l is equal to one, and $\mathcal{R}_q \cong \prod_{i=1}^{\varphi} \mathbb{Z}_q$. This is the case when $q = 1 \bmod f$, where f is a conductor of a cyclotomic, so that there are roots of unity of order f in $\mathbb{Z}_q$. Fully splitting primes are widely used in basic schemes such as ML-KEM [SAB⁺22], ML-DSA [LDK⁺22] and FN-DSA [PFH⁺22].

Unfortunately, up until now this high-performance configuration was believed to not be possible for lattice-based proof systems. The main reason is as follows. For extracting the j -th witness column $\mathbf{w}_j$ from a successful prover, the extractor obtains two accepting transcripts whose challenge vectors $\mathbf{c}$ differ only in the j -th coordinate. Then the difference $\bar{\mathbf{v}}$ of the two corresponding amortised openings is expected to be $\bar{c}_j \mathbf{w}_j$ for the difference $\bar{c}_j$ of the two j -th challenges. In the fully-splitting case the NTT coefficients of the challenges in the residue fields $\mathbb{Z}_q$ do not have enough entropy to force all NTT coefficients of the difference $\bar{c}_j$ to be non-zero. In particular, the challenge difference $\bar{c}_j \in \mathcal{R}_q$ might not be invertible. Hence, the extractor might not be able to compute the full witness. Moreover, the sumcheck phase, which works with the same arithmetic, needs large residue fields as well so that Schwartz–Zippel and the sumcheck protocol have negligible soundness error. This also prohibits fully-splitting primes in current lattice proof systems.

As mentioned, the performance of the arithmetisation and sumcheck depends on how efficiently their arithmetic can handle the arithmetic demanded by the input statement. In the important case where the input statement performs binary arithmetic, lattice proof systems need to emulate binary arithmetic in their much larger characteristic. This requires in particular lifting the binary arithmetic to $\mathbb{Z}$ and making the modulo-2 reduction explicit. And, more severely, it requires to prove that the committed witness is in fact binary.

This raises the following question:

Can we construct a lattice-based polynomial commitment scheme that simultaneously enables:

- (i) *Fast commitments* over NTT-friendly cyclotomic rings,
- (ii) *Native binary-field evaluation*, without explicit modulo-two reductions or separate bitness checks,
- (iii) *Efficient proofs of standard hash computations*, with compact proofs through recursive compression?

1.1 Our Contributions

We answer this question in the affirmative. The contributions in this work are summarised as follows:

1. We propose a new lattice-based polynomial commitment scheme (PCS) where the arithmetic of the evaluation is decoupled from that of the commitment. This allows us to instantiate the commitment in a configuration that offers superior performance over previous commitment instantiations in lattice proof systems. Moreover, it allows the evaluation proof to work over binary extension fields without any overhead. The core technical contribution lies in an extraction strategy that avoids reduction

modulo the commitment modulus q and amounts to an extraction over the number field.

2. We provide a concrete instantiation of the binary PCS that is primarily optimised for maximum prover efficiency, and secondarily tuned for small proof sizes. We combine our binary PCS with several state-of-the-art arithmetisation front-ends. To this end, we provide a field-switch reduction that carries evaluation claims over the more standard $\mathbb{F}_{2^{128}}$ to our binary extension field $\mathbb{F}_{2^{162}}$.
3. We have implemented our PCS and introduce new optimisation techniques on AVX-512 platforms. We use our implementation for proving various standard hash-functions. Our preliminary results show that lattice-based proof systems can compete with state-of-the-art code-based proof systems such as Binius [Irr26, DP25, DP26] and Flock [BRW26].
4. We explain how our PCS can easily be combined with the LaBRADOR proof system [BS23] in order to compress the proof size. This yields the smallest currently known quantum-safe proofs for standard hash functions.

1.2 Related Work

Lattice-based arguments and commitments. A central problem in lattice-based proof systems has been to establish that the committed witness has small norm or is otherwise binding. One line of work, including constructions of [AL21, CLM23, CMNW24, AFLN24, KLN024], controls norm growth during extraction through challenge sets with short inverses of pairwise differences. Another uses random projections [LNP22, NS24, KLN025, KLOT25, KLN⁺26, BBL⁺26b] or range-proofs [DBK⁺26, NOZ26] to certify shortness. The same or very similar norm-control techniques also appear in lattice-based folding, e.g. LatticeFold and LatticeFold+ [BC25a, BC25b, GLLO26] use range proofs and decomposition, while [Osa26a, KLOT25, Osa26b] rely on random projections. When the language domain does not require the extracted witness to be short, these norm-control techniques can be bypassed on the initial levels [NS24, KLN⁺26], however the advocated extraction strategy of those protocol still requires (at least statistical) invertibility of challenge differences, which constrains the choice of the commitment modulus. Except of trivial attempts of proving statements with small modulus over larger modulus with shortness guarantees (cf. [LSS24, BBL⁺26a]), none of those protocols attempted to decouple the evaluation or constraint domain from the commitment module.

Sole exceptions are CELPC [HSS24] and Jindo [HLSS26]. They commit to elements of a field $\mathbb{Z}_p$ with $p = b^\gamma + 1$ by encoding each element as γ ring coefficients of size about b , via the $\mathbb{Z}$ -module isomorphism $\mathcal{R}/(x^{\varphi/\gamma} - b) \cong \mathbb{Z}_p^{\varphi/\gamma}$. Nevertheless, this algebraic trick is applicable only to prime fields and is not transferable to binary fields.

Extension fields and general-purpose front ends. A related observation is made in Hachi [NOZ26], which reduces evaluation claims over extension fields to cyclotomic-ring statements of the same characteristic. Akita [DBK⁺26] also supports prime fields and extensions, and its integration into Jolt provides a lattice-based route to proving general RISC-V computations [DBK⁺26, GTT26]. Jolt [AST24] can express bitwise operations through lookups, and SuperSpartan [STW23] supports general algebraic constraint systems. Such front ends allow a PCS over an odd-characteristic field to prove binary computations, but the resulting arithmetisation still represents the bitwise operations in that field.

Hash-based commitments and binary proof systems. Code-based polynomial commitments provide a route to flexible evaluation fields. Brakedown [GLS⁺23], Blaze [BCF⁺25],

Bolt [GNR26], and Lightning [QGZ26] improve prover efficiency through fast encodings, while BaseFold [ZCF24] and WHIR [ACFY25] provide polylogarithmic verification. Ligerito [NA25] and Quasar [JLX⁺26] further develop this approach with different choices of codes and reductions. The strong asset of hash-based schemes is that they are often field-agnostic, i.e. relying solely on standard field properties, they do not require any adaptation of the commitment scheme to the desired field, including binary fields. The price is paid in proof size. Soundness of a hash-based commitment relies on random spot checks of a Merkle-committed codeword. At regular levels of security this amounts to a few hundred authenticated openings per proof, and proofs of a few hundred KiB even for the most compact schemes.

Arithmetisations over binary fields. At the arithmetisation level, Binius [DP25, Irr26], Flock [BRW26], and BinarySpartan [Set26] target computations with binary structure, including standard hash functions. Practically, those arithmetisations turn out to be extremely efficient due to the efficiency of binary field operations and of sumcheck over them. Concretely, as of today, these proving front ends incur an overhead of only around two orders of magnitude compared to native CPU hash execution. An immediate application of such a large throughput is the emerging need for the post-quantum transition of the Ethereum blockchain [eth26]. Two components of Ethereum rely on quantum-vulnerable cryptography, the Verkle state commitments planned for stateless clients and the BLS signatures of the consensus layer. The proposed replacements for both are hash-based, so verifying them reduces to evaluating a hash function many times. A stateless client holds only the state root and checks a block against a witness of Merkle branches for every state access, and over a binary hash tree this witness reaches about 10 MB per block, too large to broadcast every slot. A proof that the branches exist and hash to the root replaces the witness by a few hundred kilobytes, and the prover has to hash about 660 000 branch nodes in the worst case. Since the proof must be ready within a few seconds of the 12 second slot, a prover rate of about 200 000 hashes per second is estimated to suffice [But24]. Aggregating hash-based signatures [DKKW25] of the roughly 28 000 attestations per slot inside a succinct proof costs a rate of the same order, and both figures appear to be within the reach of binary-field provers on a single machine [BRW26, Set26].

The choice of arithmetisation also matters for lattice-based proof systems. To our knowledge, neither RoKoko nor Greyhound has been benchmarked on standard hash function evaluations, and their implementations use fixed finite fields whose characteristics have bit lengths between 30 and 50 [KLN⁺26, NS24]. Expressing the bitwise operations of standard hashes over such fields can introduce substantial overhead compared with arithmetisations over binary fields [DP25].

For Keccak-256, Akita [DBK⁺26] provides a lattice-based proving route through its integration with Jolt [DBK⁺26, GTTZ26], which includes an optimised Keccak-256 implementation [Jol]. A 64-byte Keccak-256 input requires one permutation [BDH⁺]. For such inputs, extrapolating from Jolt’s published cycle counts and LatticeJolt’s reported proving rates gives rough estimates of 600 hashes/s on a MacBook CPU and 3,000 hashes/s with GPU acceleration [Jol, GTTZ26].¹ The cost of expressing Keccak-256 in the zkVM must therefore be included when comparing against dedicated binary-field arithmetisations.

Proof-friendly hash functions. Arithmetisation-oriented hashes such as Rescue and Rescue-Prime [AAB⁺20, SAD20], Poseidon and Poseidon2 [GKR⁺21, GKS23], Anemoi [BBC⁺23],

¹Jolt’s optimised Keccak-256 benchmark uses 805,298 cycles to hash a 32 KiB input [Jol]. With a sponge rate of 136 bytes [BDH⁺], this requires $\lceil 32,768/136 \rceil + 1 = 241$ permutations, including padding. LatticeJolt reports over 2 million RV64IMAC cycles/s on a MacBook CPU and over 10 million cycles/s with GPU acceleration [GTTZ26]. Using these reference rates gives $241 \cdot (2 \cdot 10^6) / 805,298 \approx 600$ and $241 \cdot 10^7 / 805,298 \approx 3,000$ permutations/s. Since each 64-byte Keccak-256 hash uses one permutation [BDH⁺], these provide rough estimates for short-message hash throughput.

Griffin [GHR⁺23], and Arion [CRST26] reduce the cost of proving hash evaluations by designing the primitive around field arithmetic. A related approach in the lattice setting is developed in [BBL⁺26a], which constructs a collision-resistant hash from SIS and proposes a cryptographic hash candidate with an additional mixing component.

Cryptanalysis of these designs further motivates efficient proofs of standard hashes. Algebraic attacks have repeatedly improved on the security estimates of such designs, from the break [ACG⁺19] of Jarvis and Friday [AD18] shortly after their proposal, and distinguishers on the StarkWare-challenge instances of Poseidon [BCD⁺20], to attacks on reduced-round Poseidon, Feistel-MiMC and Rescue-Prime [BBLP22]. Gröbner-basis and resultant-based attacks [BBL⁺24, YZY⁺24, BBB⁺25, BBHN26] lower the estimated security of several parameter sets of Anemoi, Griffin, Rescue, and the original specification of Arion [RST23], whose revised specification [CRST26] increases the number of rounds in response. Attacks on Poseidon and Poseidon2 have solved reduced-round challenge instances [ZD25, BBB⁺26]. These results concern specific parameters and attack settings. Efficient binary proof systems make it possible to use standard hashes without selecting a new primitive solely for its algebraic proof cost, which is a direction also advocated in [Dra26].

1.3 Technical Overview

Throughout, let $\mathcal{R} = \mathbb{Z}[x]/(\Phi_f)$ be the ring of integers in a cyclotomic number field $\mathcal{K} = \mathbb{Q}[x]/(\Phi_f)$ of conductor f and degree $\varphi = \varphi(f)$, where $\varphi(\cdot)$ is Euler’s totient function. We use two rational moduli: a commitment modulus q , which need not be prime, and an evaluation prime p . We write $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ and $\mathcal{R}_p = \mathcal{R}/p\mathcal{R}$, and choose p inert in $\mathcal{R}$ so that $\mathcal{R}_p$ is a field.

1.3.1 Decoupling the commitment and evaluation proof algebra

Existing lattice-based proof systems use the commitment ring $\mathcal{R}_q$ also for arithmetisation and sumcheck. This couples two different requirements: the commitment modulus should support efficient arithmetic and SIS hardness, whereas the evaluation field should suit the computation being proved and provide sufficient soundness. Our approach separates these choices. We commit to a short matrix $\mathbf{W} \in \mathcal{R}^{m \times r}$ through $\mathbf{Y} = \mathbf{F}\mathbf{W} \bmod q$, where $\mathbf{F}$ is a public uniformly random matrix over $\mathcal{R}_q$, but perform arithmetisation and sumcheck modulo a different prime p . The central challenge is to ensure that these two parts of the proof refer to the same witness, despite operating in different characteristics.

The connection is provided by the evaluation and opening proof. Write $\mathbf{w}_i$ for the i -th column of $\mathbf{W}$ and $\mathbf{y}_i$ for the i -th column of $\mathbf{Y}$, so that $\mathbf{y}_i = \mathbf{F}\mathbf{w}_i \bmod q$. The evaluation claim concerns the multilinear extension of the coefficients of $\mathbf{W}$, whose variables split into row variables, indexing within a column, and column variables, indexing the columns. Let $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ be the vectors of multilinear interpolation coefficients at the row and column part of the evaluation point, so that the claimed value is $\boldsymbol{\alpha}^\top \mathbf{W} \boldsymbol{\beta}$. For the evaluation proof the prover first sends a partial evaluation where all row variables that interpolate between the rows of $\mathbf{W}$ are filled. It means the prover sends a row vector $\mathbf{u}^\top = \boldsymbol{\alpha}^\top \mathbf{W}$ that is called the *left opening*. This partial evaluation can be completed by the verifier to obtain the full evaluation $\mathbf{u}^\top \boldsymbol{\beta} = \boldsymbol{\alpha}^\top \mathbf{W} \boldsymbol{\beta}$. At the same time the verifier can check $\mathbf{u}^\top$ by testing consistency with the amortised right opening $\mathbf{v} = \mathbf{W}\mathbf{c} = \sum_i c_i \mathbf{w}_i$, where $\mathbf{c} = (c_i)_i$ is a short challenge vector, in the ring over which the evaluation is checked,

$$\mathbf{u}^\top \mathbf{c} \stackrel{?}{=} \boldsymbol{\alpha}^\top \mathbf{v}.$$

Now we see why it seems that the evaluation proof and hence the arithmetisation and sumcheck have to be instantiated over the commitment ring $\mathcal{R}_q$. The amortised opening $\mathbf{v}$

lives over $\mathcal{R}_q$ as it is used to open the commitment $\mathbf{Y}$ over $\mathcal{R}_q$, which in turn means the left opening and partial evaluation can only be checked over $\mathcal{R}_q$.

A first observation when trying to break this connection is that in the honest execution the witness vectors $\mathbf{w}_i$, as well as the challenge polynomials c_i , are short, and so in the computation of $\mathbf{v}$ there is no wrap-around modulo q . Hence, the prover effectively computes $\mathbf{v}$ over the ring of integers $\mathcal{R}$. The problem is that we do not know that the prover is honest. In fact, that $\mathbf{v}$ is short does not necessarily imply that all the $\mathbf{w}_i$ are short. A simple counterexample for power-of-two cyclotomics $\mathbb{Z}_q[x]/(x^\varphi + 1)$ is given by a coefficient w of some $\mathbf{w}_i$ whose coefficients are all $\pm(q-1)/2$ and a challenge c_i that has an even number of nonzero coefficients in $\{-1, 1\}$. Note that challenge polynomials are usually sampled to be low-weight ternary.

A natural idea to circumvent this problem is to require the prover to provide a proof that all $\mathbf{w}_i$ are in fact short. But, first, such a range proof is expensive, and second, the range proof needs to be verified over $\mathcal{R}_q$. Our goal, however, is specifically to not verify witness relations over $\mathcal{R}_q$, so that does not get us far.

Instead, we propose the following technique. Recall how a witness is extracted from amortised openings. The extractor forks the prover on the j -th coordinate of the challenge $\mathbf{c}$ and obtains two accepting right openings $\mathbf{v}$ and $\mathbf{v}'$ for challenges that differ only in c_j . Subtracting them gives $\bar{\mathbf{v}} = \mathbf{v} - \mathbf{v}'$ and $\bar{c}_j = c_j - c'_j$ with $\mathbf{F}\bar{\mathbf{v}} = \bar{c}_j \mathbf{y}_j \bmod q$, and for the honest prover $\bar{\mathbf{v}} = \bar{c}_j \mathbf{w}_j^*$ over $\mathcal{R}$. The extractor then divides by $\bar{c}_j$ to obtain a candidate witness $\mathbf{w}_j^*$ with $\bar{\mathbf{v}} = \bar{c}_j \mathbf{w}_j^*$. We do not let this division be performed in $\mathcal{R}_q$, but instead over the number field $\mathcal{K}$. Notice that the non-zero $\bar{c}_j$ is always invertible in the field $\mathcal{K}$.

In the honest execution this still gives the correct $\mathbf{w}_j$. But of course the objection from before does not go away. The prover can still commit to long vectors. Concretely, in the example above, we are now extracting a witness vector that is a proper rational vector with twos in the denominators. Moreover, the extracted vector might not even be a preimage for the commitment. We do not know if it has a prime ideal above q in the denominator and so its reduction mod q is not well-defined. But it turns out this is all fine and we can prove two properties:

- i) The commitment is binding with respect to extracted rational vectors.
- ii) The reduction modulo the prime p which we want to use for the arithmetisation is well-defined.

So why is a (sub-)commitment $\mathbf{y}_j$ binding with respect to the extracted rational vectors although $\mathbf{y}_j$ was computed over $\mathcal{R}_q$ and its opening is checked over $\mathcal{R}_q$? Observe that what we get from the extraction is a candidate *rational* opening $\mathbf{w}_j^*$ over the number field $\mathcal{K}$, together with a polynomial $\bar{c}_j \in \mathcal{R}$ that is both integral and short. Moreover, we have that $\bar{\mathbf{v}} = \bar{c}_j \mathbf{w}_j^*$ is also both integral and short, and

$$\bar{c}_j \mathbf{y}_j = \mathbf{F}\bar{\mathbf{v}} \bmod q.$$

Now suppose there is a second rational opening $\mathbf{w}_j^{*'} over $\mathcal{K}$ that also comes with an integral short $\bar{c}'_j \in \mathcal{R}$ such that $\bar{\mathbf{v}}' = \bar{c}'_j \mathbf{w}_j^{*'}$ is integral and short. Similarly to previous extraction arguments we cross multiply and subtract, which yields an element in the kernel modulo q ,$

$$\mathbf{F}(\bar{c}'_j \bar{\mathbf{v}} - \bar{c}_j \bar{\mathbf{v}}') = 0 \bmod q.$$

Next, SIS-hardness modulo q (!) implies that $\bar{c}'_j \bar{\mathbf{v}} - \bar{c}_j \bar{\mathbf{v}}'$ must be zero modulo q . Now, the central insight of our technique is that because everything in the last expression is short and known by the extractor, it actually holds over $\mathcal{R} \subset \mathcal{K}$; there is no wrap-around mod q . So we can substitute our expression for the $\bar{\mathbf{v}}$ over $\mathcal{K}$, and get

$$\bar{c}'_j \bar{c}_j \mathbf{w}_j^* - \bar{c}_j \bar{c}'_j \mathbf{w}_j^{*' = \bar{c}'_j \bar{c}_j (\mathbf{w}_j^* - \mathbf{w}_j^{*'}) = 0 \quad \text{over } \mathcal{K}.$$

Finally, because the challenge differences are invertible in $\mathcal{K}$, it holds that $\mathbf{w}_j^{*'} = \mathbf{w}_j^*$.

So, we now have succeeded in freeing the extracted vectors from the mod q restriction. In Section 3 this candidate appears as a *relaxed opening* $(\mathbf{W}, \mathbf{s})$ with $\mathbf{FW} = \mathbf{Y} \text{diag}(\mathbf{s}) \bmod q$: column j of $\mathbf{W}$ is the integral vector $\bar{\mathbf{v}}$, the *slack* s_j is the challenge difference $\bar{c}_j$, and $\mathbf{w}_j^*$ is their quotient over $\mathcal{K}$. Extending the extraction to the left opening gives, for the j -th entry u_j of $\mathbf{u}$,

$$u_j \bar{c}_j = \boldsymbol{\alpha}^\top \bar{\mathbf{v}}.$$

Normally, the consistency check is performed over $\mathcal{R}_q$ by the verifier as well, and hence the last equation also only holds over $\mathcal{R}_q$. Instead we let the consistency be checked modulo p over $\mathcal{R}_p$, and the prover also computes $\mathbf{u}^\top = \boldsymbol{\alpha}^\top \mathbf{W}$ modulo p . Next, we want to substitute $\bar{\mathbf{v}} = \bar{c}_j \mathbf{w}_j^*$ again, but need to make sure that this makes sense modulo p . In the case where $\bar{c}_j$ is invertible modulo p , i.e. it is coprime to p , the valuation of $\mathbf{w}_j^*$ is non-negative at all the prime ideals above p since $\bar{\mathbf{v}}$ is integral. So in this case the reduction of $\mathbf{w}_j^*$ modulo p is well-defined. We ensure this invertibility by choosing the challenge set so that reduction modulo p is injective. In particular, if $\mathcal{R}_p$ is a field, any two distinct challenges then have an invertible difference modulo p . For example, the elements whose coefficients in the power basis lie in $\{0, \dots, p-1\}$ map bijectively onto $\mathcal{R}_p$; a suitable subset can be used to control challenge norms. This invertibility condition comes essentially for free: we already choose $\mathcal{R}_p$ to be a sufficiently large field for sumcheck and polynomial identity testing, and injectivity of the challenge set supplies the remaining condition. No additional invertibility requirement is imposed on the commitment modulus q .

We briefly mention that the previous paragraph hints at a slightly more general observation. If the challenges have high min-entropy modulo some prime ideal $\mathfrak{p}$ and in particular if they are almost uniformly random modulo $\mathfrak{p}$, then the extracted rational opening $\mathbf{w}_j^*$ will not have a negative valuation at $\mathfrak{p}$. Put differently, negative valuations will only occur at prime ideals of small norm.

In this work, we focus solely on $p = 2$, as this regime is relevant to the main application of this work, which is to prove the execution of standard hash functions. The extracted witness over $\mathcal{K}$ need not itself have binary coefficients: its reduction modulo 2 defines the binary coefficients to which the arithmetised statement refers. We therefore avoid a separate proof that the committed witness is binary. To obtain a large binary residue field, we choose a power-of-three cyclotomic in which 2 is inert. Concretely, $\mathcal{R} = \mathbb{Z}[x]/(x^{162} + x^{81} + 1)$ gives $\mathcal{R}_2 \cong \mathbb{F}_{2^{162}}$. The same extraction strategy applies to other choices of p and $\mathcal{R}$ satisfying these conditions.

1.3.2 Efficient commitments and folding

Freedom in selecting the commitment modulus q allows us to choose a product of small fully-splitting primes instead of a single large prime. We do not stop at this, but instead further structure the commitment matrix $\mathbf{F}$ over a larger superring $\mathcal{S} \supset \mathcal{R}$. For our concrete instantiation, we choose $\mathcal{S} \cong \mathbb{Z}[x]/(x^{648} - x^{324} + 1)$ of conductor $2^3 \cdot 3^5$. Such a ring is particularly favourable for the NTT implementation, as it allows for radix-2 and radix-3 splittings. There are plenty of small primes that are NTT-friendly in $\mathcal{S}$, including 3889 and 9721, which fit in 14 bits, 17497 and 19441, which fit in 15 bits, and 2917, 4861, and 12637, which fit in 14 bits but split only into quadratic factors. These primes exhibit various tradeoffs, but all enjoy highly efficient NTT implementations. We use two or three of these primes in the product for the commitment modulus q to obtain the target SIS security level. Furthermore, since the witness is short (here, binary), folding the witness matrix $\mathbf{W}$ is done over only one fully-splitting prime. The coefficient bounds ensure that the folded result can be recovered without wraparound. Therefore, even though both commitment and folding require a single pass over the witness matrix, in practice folding is three to four times faster than commitment.

1.3.3 Controlling precisely the security level

With the ring dimensions and norm bounds fixed, restricting q to products of small primes makes security tuning coarse: adding or removing one prime changes a modulus substantially. We obtain finer control using *bit-dropping*, inspired by public-key compression in Dilithium [LDK⁺22] and adapted in [LNP22]: we omit low-order bits of the commitment and let the verifier tolerate the resulting bounded error. This reduces commitment size at the cost of increasing the norm bound in the underlying SIS problem, allowing us to trade excess security for smaller proofs without changing the modulus.

1.3.4 Recursion with LaBRADOR

A significant advantage of our protocol, and lattice-based proofs in general, is that, due to the simple algebraic structure of the commitment, the evaluation proof verifier only has to perform cyclotomic ring arithmetic apart from Fiat-Shamir hashes. This allows for straight-forward recursive proofs of the verifier with a further lattice proof system. We exemplify this using LaBRADOR [BS23]. More precisely, for a witness containing $N = mr$ ring elements, balancing the matrix dimensions gives a commitment and openings containing $O(\sqrt{N})$ ring elements. To get further compression, we wrap all prover communication (except the sumcheck messages) in LaBRADOR, which folds the commitments and openings of our scheme and proves the consistency of the commitments and openings with respect to the verifier checks. Its prover and verifier costs are linear in this reduced instance, whose size is proportional to $\sqrt{N}$. Such compression, although concretely expensive for the verifier, gives a substantial reduction in proof size to essentially constant size of $\approx 80\text{KB}$.

1.3.5 Integration with Binius and Flock

We integrate our PCS with the PIOPs underlying Binius [Irr26, DP25, DP26] and Flock [BRW26] to prove standard hash computations. These protocols produce evaluation claims over $\mathbb{F}_{2^{128}}$, whereas our PCS operates over $\mathbb{F}_{2^{162}}$. A field-switch reduction inspired by [DP26] connects the two without modifying the front-end arithmetisation. We pack the 128 binary coordinates of each source-field element into an $\mathbb{F}_2$ -subspace of $\mathbb{F}_{2^{162}}$. To connect the evaluation claims, the prover sends partial evaluations whose recombination in $\mathbb{F}_{2^{128}}$ must equal the claimed value. The verifier then randomly aggregates their packed versions, and a sumcheck over $\mathbb{F}_{2^{162}}$ reduces their consistency with the packed witness to a single evaluation claim for our PCS.

The resulting systems retain the front-end constraint reductions while replacing their commitment and opening layers with our Ajtai commitment and evaluation proof. Our AVX-512 benchmarks in Section 5 show competitive non-recursive prover and verifier times, while the LaBRADOR wrapper produces substantially smaller proofs than the compared Binius and Flock configurations. In the most favourable non-recursive configuration, we prove over 67,000 BLAKE3 compression calls per second on a single CPU core, corresponding to a throughput of about 4.3 MB of data per second.

1.4 Perspective

Our construction shifts attention to the cost of recursive compression. Wrapping our PCS in LaBRADOR [BS23] gives small proofs, but leaves room to improve verification time. A natural next step is to explore a wrapper based on RoKoko [KLN⁺26], or a dedicated argument tailored to the verifier of our PCS. Such a construction could retain the benefits of our commitment and evaluation protocols while reducing the cost of verifying the final proof.

A second direction is to extend our construction to zero knowledge. A possible approach is to hide the commitments and openings inside a zero-knowledge argument such

as LaBRADOS [BBL⁺26a], and to blind the sumcheck messages using techniques similar to those of [XZZ⁺19]. Turning this approach into an efficient construction requires ensuring that the entire composed protocol hides the witness, including the information revealed by the evaluation proof.

A longer-term application of our work is incrementally verifiable computation (IVC) [Val08, KST22], where each step certifies both a new computation and the validity of the preceding proof. Recent lattice-based folding schemes provide promising building blocks for this task [BC25b, BC25a, KLOT25, NS25, NS26, GLLO26, Osa26a, Osa26b]. Their use of Ajtai commitments parallels the role of Pedersen commitments [Ped92] in discrete-log-based folding schemes: the verifier can form a commitment to a linear combination of witnesses by taking the same linear combination of their commitments, without accessing the witnesses themselves. This homomorphism is what allows the commitments to be folded alongside the witnesses. Turning these folding schemes into IVC also requires certifying the hash computations used by the Fiat-Shamir transform, which are not directly captured by this algebraic folding. Our construction brings these two requirements together: it supports efficient proofs of standard hash computations while retaining an Ajtai commitment to the witness. Realising this integration remains an open direction.

2 Preliminaries

Let $\mathbb{N} := \{1, 2, \dots\}$. For $n \in \mathbb{N}$, define $[n] := \{0, \dots, n-1\}$. For $i \in [2^\nu]$, write $\text{bin}(i) = (i_0, \dots, i_{\nu-1}) \in \{0, 1\}^\nu$ for its little-endian binary expansion, so that $i = \sum_{j \in [\nu]} 2^j i_j$. We identify $\{0, 1\}^\nu$ with $[2^\nu]$ through bin , and vectors indexed by $\{0, 1\}^\nu$ order their entries by the little-endian value of the index. The notation $x \leftarrow \$ S$ denotes uniform sampling from a finite set S unless a different distribution is specified. For an integer $q \geq 2$, write

$$[\pm q/2] := \{-\lceil q/2 \rceil + 1, \dots, \lfloor q/2 \rfloor\}.$$

We identify every element of $\mathbb{Z}_q$ with its unique representative in $[\pm q/2]$. Reduction modulo q of an integer vector or matrix always returns the coordinate-wise balanced representative.

Bold lower-case letters denote vectors and bold upper-case letters denote matrices. The superscript $\top$ denotes transpose, $\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{x}^\top \mathbf{y}$ is the standard bilinear inner product, taken coefficientwise after zero extension to the larger length when the two vectors differ in length, and $\otimes$ denotes the Kronecker product. All logarithms are base two. The security parameter is $\lambda \in \mathbb{N}$, and public parameters implicitly contain 1^λ .

2.1 Cyclotomic Rings with Two Inert

Contrary to the usual convention in lattice-based cryptography, we work over a cyclotomic ring of odd conductor, chosen so that 2 is inert and reduction modulo 2 maps the ring onto a large binary field.

Definition 1 (Cyclotomic Setting). Let f be an odd conductor for which 2 is a primitive root modulo f , i.e. $\text{ord}_f(2) = \varphi(f) = \varphi$, where φ is Euler's totient function. Let $\mathcal{K} := \mathbb{Q}(\zeta_f)$ be the f -th cyclotomic number field, where ζ_f is a primitive f -th root of unity, and let $\mathcal{R} := \mathbb{Z}[\zeta_f]$ be its ring of integers, a free $\mathbb{Z}$ -module with power basis $(\zeta_f^i)_{i \in [\varphi]}$. For an integer $q \geq 2$, let $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$. We identify every element of $\mathcal{R}_q$ with the unique element of $\mathcal{R}$ whose coefficient vector over the power basis is balanced coordinate-wise, and all norms of an element of $\mathcal{R}_q$ are those of this lift.

Definition 2 (Norms over $\mathcal{R}$). Every $x \in \mathcal{R}$ has a unique coefficient vector $\text{coeff}(x) \in \mathbb{Z}^\varphi$ with $x = \sum_{i \in [\varphi]} \text{coeff}(x)_i \zeta_f^i$, and we set $\|x\| := \|\text{coeff}(x)\|_2$. For a vector $\mathbf{v} \in \mathcal{R}^n$ we set

$\|\mathbf{v}\|^2 := \sum_{i \in [n]} \|v_i\|^2$, and for a matrix $\mathbf{W} \in \mathcal{R}^{n \times r}$ we take the column-wise maximum $\|\mathbf{W}\| := \max_{j \in [r]} \|\mathbf{W}_{:,j}\|$. The operator norm of $x \in \mathcal{R}$ is

$$\|x\|_{\text{op}} := \sup_{y \in \mathcal{R} \setminus \{0\}} \frac{\|xy\|}{\|y\|},$$

so that $\|xy\| \leq \|x\|_{\text{op}} \|y\|$ for all $y \in \mathcal{R}$.

Two elements of $\mathcal{R}$ of norm below $q/2$ that agree over $\mathcal{R}_q$ are equal over $\mathcal{R}$, since every coefficient of their difference is a multiple of q of absolute value below q .

Lemma 1 (Inertness of Two). *In the setting of Definition 1, the ideal $2\mathcal{R}$ is prime in $\mathcal{R}$, and $\mathcal{R}/2\mathcal{R}$ is the field with 2^φ elements.*

Proof. Since $\mathfrak{f}$ is odd, $2 \nmid \mathfrak{f}$, so 2 is unramified in $\mathcal{K}$. A rational prime $p \nmid \mathfrak{f}$ factors in $\mathcal{R}$ into φ/d distinct prime ideals of residue degree $d = \text{ord}_{\mathfrak{f}}(p)$, or equivalently $\Phi_{\mathfrak{f}}$ factors modulo p into φ/d irreducible factors of degree d . For $p = 2$, the hypothesis $\text{ord}_{\mathfrak{f}}(2) = \varphi$ forces a single factor, so $2\mathcal{R}$ is prime and $\mathcal{R}/2\mathcal{R}$ is the field with 2^φ elements. $\square$

Remark 1 (Powers of Three). The hypothesis of Definition 1 holds for every $\mathfrak{f} = 3^k$ with $k \geq 1$. Indeed, 2 is a primitive root modulo 3, and $2^{3-1} \bmod 9 = 4 \neq 1$, so by the standard lifting criterion for primitive roots, 2 is a primitive root modulo 3^k for all $k \geq 1$. Hence $\mathcal{R}/2\mathcal{R}$ is the field with $2^{\varphi(3^k)} = 2^{2 \cdot 3^{k-1}}$ elements.

Definition 3 (Binary Residue Field). Let $\mathbb{F} := \mathcal{R}_2 = \mathcal{R}/2\mathcal{R}$, which by Lemma 1 is the binary field of degree φ . For $x \in \mathcal{R}$, we write $x \bmod 2 \in \mathbb{F}$ for its image under the reduction $\mathcal{R} \rightarrow \mathbb{F}$, applied entry-wise to vectors and matrices. Since $\mathbb{F}$ is a field, $x \bmod 2$ is invertible exactly when $x \notin 2\mathcal{R}$.

2.2 The SIS Assumption

We state the short integer solution (SIS) problem over $\mathcal{R}$ [Ajt96].

Definition 4 (SIS). Let $m, n \in \mathbb{N}$ and $\beta \in \mathbb{R}$. Define the relation

$$\Xi_{\mathcal{R},q,m,n,\beta}^{\text{SIS}} := \left\{ (\mathbf{F}, \mathbf{x}) : \begin{array}{l} \mathbf{F} \in \mathcal{R}_q^{n \times m}; \mathbf{x} \in \mathcal{R}^m \\ \|\mathbf{x}\| \leq \beta; \mathbf{F}\mathbf{x} = 0 \bmod q; \mathbf{x} \neq 0 \end{array} \right\}.$$

The $\text{SIS}_{\mathcal{R},q,m,n,\beta}$ problem is, given $\mathbf{F} \leftarrow \mathcal{R}_q^{n \times m}$, to find $\mathbf{x}$ with $(\mathbf{F}, \mathbf{x}) \in \Xi_{\mathcal{R},q,m,n,\beta}^{\text{SIS}}$.

Definition 5 (SIS Hardness). For parameters depending on the security parameter λ , the $\text{SIS}_{\mathcal{R},q,m,n,\beta}$ problem is *hard* if, for every probabilistic expected polynomial-time adversary $\mathcal{A}$,

$$\Pr [(\mathbf{F}, \mathbf{x}) \in \Xi_{\mathcal{R},q,m,n,\beta}^{\text{SIS}} \mid \mathbf{F} \leftarrow \mathcal{R}_q^{n \times m}, \mathbf{x} \leftarrow \mathcal{A}(\mathbf{F})] \leq 2^{-\lambda}.$$

2.3 Challenge Sets and Coordinate-Wise Extraction

Extraction for the folding protocol divides the recovered witness by differences of distinct challenges, but only modulo 2. The requirement on the challenge set is therefore that distinct challenges remain distinct modulo 2.

Definition 6 (Folding Challenge Sets). Let $\mathcal{C} \subseteq \mathcal{R}$ be finite. Its expansion factor is

$$\gamma_{\mathcal{C}} := \max_{c \in \mathcal{C}} \|c\|_{\text{op}}.$$

The set $\mathcal{C}$ is *injective modulo 2* if reduction modulo 2 is injective on $\mathcal{C}$, that is,

$$c - c_0 \notin 2\mathcal{R} \quad \text{for all } c, c_0 \in \mathcal{C} \text{ with } c \neq c_0.$$

We will use the following lemma to construct folding challenge sets that are injective modulo 2. Flipping the sign of a coefficient makes the elements of the challenge set approximately zero-centered, which is convenient for the average-case analysis of the norm growth.

Lemma 2 (Signed Binary Challenges). *Let $\mathcal{B} := \left\{ \sum_{i \in [\varphi]} b_i \zeta_i^i : \mathbf{b} \in \{0, 1\}^\varphi \right\}$. Every subset $\mathcal{C} \subseteq \mathcal{B}$ is injective modulo 2. For a public hash function $H: \mathcal{R} \rightarrow \{-1, 1\}^\varphi$, define $s_H(c) := \sum_{i \in [\varphi]} H(c)_i b_i \zeta_i^i$ for $c = \sum_{i \in [\varphi]} b_i \zeta_i^i \in \mathcal{B}$. Then $s_H(\mathcal{C})$ is injective modulo 2, and every element of $s_H(\mathcal{C})$ has coefficients in $\{-1, 0, 1\}$.*

Proof. Reduction modulo 2 maps $\mathcal{B}$ bijectively onto $\mathbb{F}$, so it is injective on $\mathcal{C}$. Since $-1 = 1 \pmod 2$, we have $s_H(c) = c \pmod 2$ for every $c \in \mathcal{B}$. Thus distinct elements of $\mathcal{C}$ have distinct signed images modulo 2. The final claim follows directly from $b_i \in \{0, 1\}$ and $H(c)_i \in \{-1, 1\}$. $\square$

2.4 Coordinate-Wise Extraction

Definition 7 (Coordinate-Wise Special-Soundness Families [FMN24]). Let $k, \ell \in \mathbb{N}$, let S be a finite set, and define $K := \ell(k-1) + 1$. The family $\mathbf{SS}(S, \ell, k)$ consists of the sets $\mathcal{C} = \{\mathbf{c}_0, \dots, \mathbf{c}_{K-1}\} \subseteq S^\ell$ of cardinality K for which there is a central index $e \in [K]$ such that, for every $i \in [\ell]$, there is a set $J_i \subseteq [K] \setminus \{e\}$ of cardinality $k-1$ satisfying, for every $j \in J_i$,

$$c_{e,i} \neq c_{j,i} \quad \text{and} \quad c_{e,h} = c_{j,h} \quad \text{for every } h \in [\ell] \setminus \{i\}.$$

Lemma 3 (Coordinate-Wise Forking [FMN24, Lemma 7.1]). *Let $k, \ell \in \mathbb{N}$, let S be a finite set of cardinality N , define $\mathcal{C} := S^\ell$, and let $V: \mathcal{C} \times \{0, 1\}^* \rightarrow \{0, 1\}$ be any verification function. There is an oracle algorithm $\mathcal{E}$ with the following properties. Given oracle access to a probabilistic algorithm $\mathcal{A}: \mathcal{C} \rightarrow \{0, 1\}^*$, the algorithm $\mathcal{E}^{\mathcal{A}}$ makes at most $\ell(k-1) + 1$ queries to $\mathcal{A}$ in expectation and, with probability at least*

$$\epsilon_V(\mathcal{A}) - \frac{\ell(k-1)}{N}, \quad \epsilon_V(\mathcal{A}) := \Pr_{\mathbf{c} \leftarrow \mathcal{C}}[V(\mathbf{c}, \mathcal{A}(\mathbf{c})) = 1],$$

outputs $\ell(k-1) + 1$ pairs $(\mathbf{c}_i, y_i)_{i \in [\ell(k-1)+1]}$ such that $V(\mathbf{c}_i, y_i) = 1$ for every $i \in [\ell(k-1)+1]$ and

$$\{\mathbf{c}_i : i \in [\ell(k-1) + 1]\} \in \mathbf{SS}(S, \ell, k).$$

2.5 Multilinear Polynomials and Sumcheck

The equality polynomial has integer coefficients, hence is defined over any commutative ring. We therefore state the multilinear machinery over an arbitrary commutative ring A , which in our application ranges over $\mathbb{F}$, $\mathbb{T}$, $\mathcal{R}$, and $\mathcal{R}_q$.

Definition 8 (Vectorisation). For a matrix $\mathbf{A} \in A^{a \times b}$, the column-major vectorisation $\text{vec}(\mathbf{A}) \in A^{ab}$ places the entry of $\mathbf{A}$ in row $i \in [a]$ and column $j \in [b]$ at index $ja + i$.

Definition 9 (Equality Polynomial). Let A be a commutative ring and $\nu \in \mathbb{N}$. The equality polynomial $\text{eq}: A^\nu \times A^\nu \rightarrow A$ is

$$\text{eq}(\mathbf{x}, \mathbf{y}) := \prod_{i \in [\nu]} (x_i y_i + (1 - x_i)(1 - y_i)),$$

and for $\mathbf{r} \in A^\nu$, its evaluation vector over the Boolean hypercube is

$$\tilde{\text{eq}}(\mathbf{r}) := (\text{eq}(\mathbf{r}, \mathbf{b}))_{\mathbf{b} \in \{0, 1\}^\nu} = (1 - r_{\nu-1}, r_{\nu-1}) \otimes \dots \otimes (1 - r_0, r_0) \in A^{2^\nu},$$

the tensor form following from the little-endian ordering fixed in Section 2.

Definition 10 (Multilinear Extension). Let A be a commutative ring, $\nu \in \mathbb{N}$, and $\mathbf{a} \in A^{2^\nu}$ indexed by the hypercube $\{0, 1\}^\nu$. The *multilinear extension* $\text{MLE}[\mathbf{a}]: A^\nu \rightarrow A$ of $\mathbf{a}$ is the multilinear polynomial

$$\text{MLE}[\mathbf{a}](\mathbf{r}) := \langle \tilde{\mathbf{eq}}(\mathbf{r}), \mathbf{a} \rangle = \tilde{\mathbf{eq}}(\mathbf{r})^\top \mathbf{a}, \quad (1)$$

which is the unique multilinear polynomial that agrees with $\mathbf{a}$ on the hypercube, i.e. $\text{MLE}[\mathbf{a}](\mathbf{b}) = a_{\mathbf{b}}$ for all $\mathbf{b} \in \{0, 1\}^\nu$.

The equality vector factors along a splitting of its variables, and the column-major vectorisation lets us evaluate the extension of a matrix as a bilinear form in its rows and columns.

Lemma 4 (Bilinear Evaluation). *Let A be a commutative ring, $\mathbf{W} \in A^{m \times r}$ with m, r powers of two, and let $\mathbf{r} \in A^{\log(mr)}$ split as $\mathbf{r}^\top = (\mathbf{r}_0^\top \ \mathbf{r}_1^\top)$ with $\mathbf{r}_0 \in A^{\log m}$ and $\mathbf{r}_1 \in A^{\log r}$. Then $\tilde{\mathbf{eq}}(\mathbf{r}) = \tilde{\mathbf{eq}}(\mathbf{r}_1) \otimes \tilde{\mathbf{eq}}(\mathbf{r}_0)$ and*

$$\text{MLE}[\text{vec}(\mathbf{W})](\mathbf{r}) = \tilde{\mathbf{eq}}(\mathbf{r}_0)^\top \mathbf{W} \tilde{\mathbf{eq}}(\mathbf{r}_1).$$

Proof. The product defining $\mathbf{eq}$ splits over the two blocks of variables, and under the little-endian ordering the first block indexes the rows and the second the columns of $\mathbf{W}$, matching the column-major vectorisation, so $\tilde{\mathbf{eq}}(\mathbf{r}) = \tilde{\mathbf{eq}}(\mathbf{r}_1) \otimes \tilde{\mathbf{eq}}(\mathbf{r}_0)$. Substituting into (1) and using $\text{vec}(\mathbf{W})^\top(\mathbf{y} \otimes \mathbf{x}) = \mathbf{x}^\top \mathbf{W} \mathbf{y}$ yields the claim. $\square$

2.6 Polynomial Identity Testing and Sumcheck

Lemma 5 (Polynomial Root Bound [BCPS18, Theorem 4.2]). *Let L be a field, let $S \subseteq L$ be finite, and let $f \in L[x_0, \dots, x_{\ell-1}]$ be a non-zero polynomial of total degree at most D . Then*

$$\Pr_{\mathbf{r} \leftarrow S^\ell} [f(\mathbf{r}) = 0] \leq \frac{D}{|S|}.$$

Definition 11 (Sumcheck Protocol). Let L be a finite field, let $\ell, d \in \mathbb{N}$, let $f \in L[x_0, \dots, x_{\ell-1}]$ have individual degree at most d , and let $t \in L$. The *sumcheck protocol* for the claim $\sum_{\mathbf{b} \in \{0, 1\}^\ell} f(\mathbf{b}) = t$ proceeds in ℓ rounds, starting from $t_0 := t$. In round $i \in [\ell]$, the prover sends the polynomial

$$g_i(\mathbf{x}) := \sum_{\mathbf{b} \in \{0, 1\}^{\ell-1-i}} f(r_0, \dots, r_{i-1}, \mathbf{x}, \mathbf{b})$$

of degree at most d , and the verifier checks $g_i(0) + g_i(1) \stackrel{?}{=} t_i$, samples $r_i \leftarrow L$, and sets $t_{i+1} := g_i(r_i)$. After the last round, the verifier outputs the claim $f(\mathbf{r}) = t_\ell$ for $\mathbf{r} := (r_0, \dots, r_{\ell-1})$.

Lemma 6 (Sumcheck Soundness [LFKN92]). *The sumcheck protocol is perfectly complete. If $\sum_{\mathbf{b} \in \{0, 1\}^\ell} f(\mathbf{b}) \neq t$, then for every prover strategy the probability, over the challenges $\mathbf{r} \leftarrow L^\ell$, that every round check passes and the output claim $f(\mathbf{r}) = t_\ell$ holds is at most $\ell d / |L|$.*

2.7 Field Switching

Since we intend to use front ends like Binius or Flock in a black-box manner, the field over which the witness is defined may differ from the field over which the commitments are defined. In particular, $\mathbb{T}$ may be a 128-bit tower field, while the committed witness lives over the binary residue field $\mathbb{F}$ of degree φ . In general, there is no field embedding $\mathbb{T} \rightarrow \mathbb{F}$, as the degree of $\mathbb{T}$ need not divide φ . The switch between the two fields therefore relies only on their common $\mathbb{F}_2$ -linear structure.

Definition 12 (Field-Switch Packing). Let $\iota \in \mathbb{N}$ with $2^\iota \leq \varphi$ and let $\mathbb{T}$ be a binary field of degree 2^ι . Fix an $\mathbb{F}_2$ -basis $(\theta_{\mathbf{k}})_{\mathbf{k} \in \{0,1\}^\iota}$ of $\mathbb{T}$, collected into $\boldsymbol{\theta} := (\theta_{\mathbf{k}})_{\mathbf{k} \in \{0,1\}^\iota} \in \mathbb{T}^{2^\iota}$, and an $\mathbb{F}_2$ -linearly independent family $(\eta_{\mathbf{k}})_{\mathbf{k} \in \{0,1\}^\iota}$ in $\mathbb{F}$, spanning $V := \text{span}_{\mathbb{F}_2} \{\eta_{\mathbf{k}}\} \subseteq \mathbb{F}$. The *field-switch packing* is the $\mathbb{F}_2$ -linear bijection $\phi: \mathbb{T} \rightarrow V$ with $\phi(\theta_{\mathbf{k}}) = \eta_{\mathbf{k}}$ for all $\mathbf{k}$. Extend $(\eta_{\mathbf{k}})_{\mathbf{k} \in \{0,1\}^\iota}$ by further elements to an $\mathbb{F}_2$ -basis of $\mathbb{F}$, and let $P_V: \mathbb{F} \rightarrow V$ be the $\mathbb{F}_2$ -linear projection that keeps the coordinates over $(\eta_{\mathbf{k}})_{\mathbf{k} \in \{0,1\}^\iota}$ and zeroes the remaining ones. Both maps are applied entry-wise to vectors.

Both $\mathbb{T}$ and $\mathbb{F}$ lift into $\mathcal{R}$ by reading their coordinates as coefficients over the power basis.

Definition 13 (Binary Lift). Let L be a binary field of degree $d \leq \varphi$ with a fixed $\mathbb{F}_2$ -basis $(\theta_i)_{i \in [d]}$: for $L = \mathbb{T}$ the basis $\boldsymbol{\theta}$ of Definition 12, and for $L = \mathbb{F}$ the reduced power basis $(\zeta_f^i \bmod 2)_{i \in [\varphi]}$. The *binary lift* $\text{lift}: L \rightarrow \mathcal{R}$ maps the coordinates over $(\theta_i)_{i \in [d]}$ to coordinates over the power basis,

$$\text{lift}\left(\sum_{i \in [d]} b_i \theta_i\right) := \sum_{i \in [d]} b_i \zeta_f^i \quad \text{for } b_i \in \{0,1\},$$

applied entry-wise to vectors and matrices.

Definition 14 (Basis Decomposition of the Equality Vector). Let $\phi: \mathbb{T} \rightarrow V$ be a field-switch packing and $\mathbf{r} \in \mathbb{T}^\nu$. The matrix $\mathbf{E}(\mathbf{r}) \in \mathbb{F}_2^{2^\iota \times 2^\nu}$ is defined by the decompositions

$$\text{eq}(\mathbf{r}, \mathbf{b}) = \sum_{\mathbf{k} \in \{0,1\}^\iota} \mathbf{E}(\mathbf{r})_{\mathbf{k}, \mathbf{b}} \theta_{\mathbf{k}} \quad \text{for all } \mathbf{b} \in \{0,1\}^\nu,$$

i.e. column $\mathbf{b}$ of $\mathbf{E}(\mathbf{r})$ holds the $\boldsymbol{\theta}$ -coordinates of the entry of $\tilde{\text{eq}}(\mathbf{r})$ at index $\mathbf{b}$. Entry-wise, the definition reads $\tilde{\text{eq}}(\mathbf{r})^\top = \boldsymbol{\theta}^\top \mathbf{E}(\mathbf{r})$ over $\mathbb{T}$.

Since $\mathbf{E}(\mathbf{r})$ has entries in $\mathbb{F}_2$, it acts on vectors over every $\mathbb{F}_2$ -vector space, in particular over both $\mathbb{T}$ and $\mathbb{F}$.

Lemma 7 (Entry-Wise Commutation). Let $\mathbf{E} \in \mathbb{F}_2^{a \times b}$, let U, U' be $\mathbb{F}_2$ -vector spaces, and let $\psi: U \rightarrow U'$ be an $\mathbb{F}_2$ -linear map, applied entry-wise to vectors. Then $\mathbf{E} \psi(\mathbf{v}) = \psi(\mathbf{E} \mathbf{v})$ for every $\mathbf{v} \in U^b$. In particular, $\mathbf{E}(\mathbf{r})$ commutes with ϕ , ϕ^{-1} , and P_V .

Proof. Entry s of $\mathbf{E} \mathbf{v}$ is the sum of the entries v_t with $E_{s,t} = 1$, and ψ preserves such sums. $\square$

2.8 Reductions of Knowledge

We recall the notion of reduction of knowledge introduced by Kothapalli and Parno [KP23].

Definition 15 (Reduction of Knowledge, Adapted from [KLNO24, KLNO25, KLOT25]). A *reduction of knowledge* (RoK) $\Pi = (\mathcal{P}, \mathcal{V})$ consists of interactive probabilistic polynomial-time algorithms with the following interface:

- $\mathcal{P}(\text{pp}, \text{stmt}_0, \text{wit}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ interactively reduces an input instance to an output instance or returns $\perp$.
- $\mathcal{V}(\text{pp}, \text{stmt}_0) \rightarrow \text{stmt}_1$ interactively reduces verification of the input statement to verification of the output statement.

We write $\langle \mathcal{P}, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0, \text{wit}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ for an honest interaction, whose output consists of the statement stmt_1 returned by $\mathcal{V}$ and the witness wit_1 returned by $\mathcal{P}$. The notation $\langle \mathcal{P}^*, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ denotes the same interaction with a malicious prover $\mathcal{P}^*$ in place of $\mathcal{P}$.

Let Ξ_0 and Ξ_1 be relations. The RoK is *correct* for $\Xi_0 \rightarrow \Xi_1$ with correctness error $\delta := \delta(\text{pp}, \text{stmt}_0)$ if, for every $(\text{pp}, \text{stmt}_0, \text{wit}_0) \in \Xi_0$,

$$\Pr[(\text{pp}, \text{stmt}_1, \text{wit}_1) \in \Xi_1 \mid (\text{stmt}_1, \text{wit}_1) \leftarrow \langle \mathcal{P}, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0, \text{wit}_0)] \geq 1 - \delta(\text{pp}, \text{stmt}_0),$$

the probability being over the randomness of $\mathcal{P}$ and $\mathcal{V}$. The correctness error is taken as $\delta = 0$ if omitted. If a RoK Π is correct for $\Xi_0 \rightarrow \Xi_1$, we say that it is a RoK from Ξ_0 to Ξ_1 , short $\Pi: \Xi_0 \rightarrow \Xi_1$.

Let Ξ'_0 and Ξ'_1 be extraction relations. The RoK is *knowledge sound* for $\Xi'_0 \leftarrow \Xi'_1$ with knowledge error $\kappa := \kappa(\text{pp}, \text{stmt}_0)$ if there is a *black-box* expected polynomial-time extractor $\mathcal{E}$ such that, for every $(\text{pp}, \text{stmt}_0)$ and every possibly unbounded malicious prover $\mathcal{P}^*$,

$$\begin{aligned} \Pr[(\text{pp}, \text{stmt}_0, \text{wit}_0) \in \Xi'_0 \mid \text{wit}_0 \leftarrow \mathcal{E}^{\mathcal{P}^*}(\text{pp}, \text{stmt}_0)] \\ \geq \Pr[(\text{pp}, \text{stmt}_1, \text{wit}_1) \in \Xi'_1 \mid (\text{stmt}_1, \text{wit}_1) \leftarrow \langle \mathcal{P}^*, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0)] \\ - \kappa(\text{pp}, \text{stmt}_0), \end{aligned}$$

the probabilities being over the randomness of $\mathcal{V}$, $\mathcal{P}^*$, and $\mathcal{E}$. Oracle calls to $\mathcal{P}^*$ have unit cost.

A proof of knowledge is the degenerate case of a reduction of knowledge in which nothing remains to be checked after the interaction.

Definition 16 (Proof of Knowledge). Let $\Xi_\top := \{(\text{pp}, \top, \varepsilon)\}$ be the relation whose only statement is the distinguished accepting statement $\top$, with the empty witness ε . A *proof of knowledge* for Ξ with extraction relation Ξ' , correctness error δ , and knowledge error κ is a RoK that is correct for $\Xi \rightarrow \Xi_\top$ with error δ and knowledge sound for $\Xi' \leftarrow \Xi_\top$ with error κ .

2.9 Polynomial Commitment Schemes

Our reductions compose into an evaluation proof for a committed multilinear polynomial. We therefore state the evaluation proof as a proof of knowledge for a relation containing the opening check, so that the scheme is extractable. Due to knowledge soundness of the proof, this is a stronger notion than the evaluation binding usually required of a polynomial commitment scheme, which only forbids accepting proofs of two different values $t \neq t'$ at the same point for the same commitment.

Definition 17 (Commitment Scheme). Let $\mathcal{M}$ be a message space. A *commitment scheme* over $\mathcal{M}$ is a tuple of probabilistic polynomial-time algorithms (**Setup**, **Com**, **Open**) with the following interface:

- **Setup**(1^λ) $\rightarrow$ **pp**: sample public parameters.
- **Com**(**pp**, f) $\rightarrow$ (c , **st**): commit to a message $f \in \mathcal{M}$, returning a commitment c and an opening state **st**.
- **Open**(**pp**, c , f , **st**) $\rightarrow$ $\{0, 1\}$: check that **st** opens c to f .

It is *complete* with completeness error δ if, for every $f \in \mathcal{M}$,

$$\Pr[\text{Open}(\text{pp}, c, f, \text{st}) = 1 \mid \text{pp} \leftarrow \text{Setup}(1^\lambda), (c, \text{st}) \leftarrow \text{Com}(\text{pp}, f)] \geq 1 - \delta.$$

Definition 18 (Binding). A commitment scheme is *binding* with binding error ϵ if, for every probabilistic polynomial-time adversary $\mathcal{A}$,

$$\Pr \left[\begin{array}{l} f \neq f' \wedge \text{Open}(\text{pp}, c, f, \text{st}) = 1 \\ \wedge \text{Open}(\text{pp}, c, f', \text{st}') = 1 \end{array} \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (c, (f, \text{st}), (f', \text{st}')) \leftarrow \mathcal{A}(\text{pp}) \end{array} \right] \leq \epsilon.$$

Definition 19 (Multilinear Polynomial Commitment Scheme). Let L be a finite field and $\nu \in \mathbb{N}$. A *multilinear polynomial commitment scheme* for ν variables over L with correctness error δ and knowledge error κ consists of a complete and binding commitment scheme (**Setup**, **Com**, **Open**) over the message space L^{2^ν} , whose messages are identified with their multilinear extensions (Definition 10), together with a proof of knowledge (Definition 16) with correctness error δ and knowledge error κ for the evaluation relation

$$\Xi^{\text{eval}} := \{(\text{pp}, (c, \mathbf{r}, t), (\mathbf{f}, \text{st})) : \text{Open}(\text{pp}, c, \mathbf{f}, \text{st}) = 1 \wedge \text{MLE}[\mathbf{f}](\mathbf{r}) = t\},$$

where $\mathbf{f} \in L^{2^\nu}$, $\mathbf{r} \in L^\nu$, and $t \in L$. The proof of knowledge is required to be correct only on tuples whose (c, st) is an output of $\text{Com}(\text{pp}, \mathbf{f})$.

3 Polynomial Commitment Scheme over Binary Field

Our construction of a polynomial commitment scheme over a binary field $\mathbb{T}$ is based on composition of the following reductions:

1. Reduction Π^{sw} from $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw}}$ to $\Xi_{\text{par}_{\text{out}}}^{\text{bin-eval}}$, which is a field-switching reduction that allows us to use a polynomial commitment scheme over $\mathbb{F} = \mathcal{R}_2$ for a polynomial commitment scheme over $\mathbb{T}$. This reduction is described in Figure 2 and Lemma 8. Such a reduction is necessary as we do not intend our scheme to operate solely on a binary extension of degree tied to the cyclotomic ring $\mathcal{R}$.
2. Reduction $\Pi^{\text{left-exp}}$ from $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval}}$ to $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$, which is a reduction from the binary evaluation relation to the linear relation with binary claims. This reduction is described in Figure 3 and Lemma 9.
3. Reduction Π^{fold} from $\Xi_{\text{par}_{\text{in}}}^{\text{bin-lin}}$ to $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$, which is a self-reduction of the linear relation with binary claims that folds the r witness columns into one. This reduction is described in Figure 4 and Lemma 10.

All of the reductions are correct and knowledge-sound, but the extraction is over *relaxed* relations, which is sufficient for our purposes. Figure 1 shows the flow of the reductions and the relations they reduce to. Those reductions are eventually composed to obtain a polynomial commitment scheme over the binary field $\mathbb{T}$ in Theorem 1.

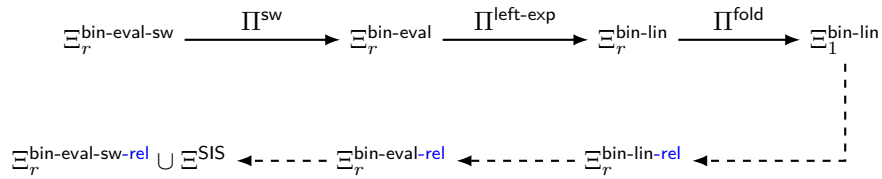


Figure 1: Correctness (solid) and extraction (dashed) flows of the reductions, with subscripts counting the witness columns.

3.1 Relations

We consider the following relations. All of the relations are defined in *exact* and *relaxed* variants. The relaxed variants are used for knowledge extraction, while the exact variants are used for correctness. The exact variant of a relation is obtained by removing the *blue* parts of its definition, i.e. by fixing the slack $\mathbf{s} = \mathbf{1}$ and dropping the parameter μ , so an exact witness is a relaxed witness for every $\mu \geq 1$. Furthermore, all relations used as protocol interfaces, i.e. $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw-rel}}$, $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-rel}}$, $\Xi_{\text{par}_{\text{in}}}^{\text{bin-lin-rel}}$, inherit from a common principal relation $\Xi^{\text{lin-rel}}$.

Definition 20 (Relaxed Principal Relation). Let $m, n, r \in \mathbb{N}$, $\beta, \mu \in \mathbb{R}$. We define the relation

$$\Xi_{\mathcal{R}, q, m, n, r, \beta, \mu}^{\text{lin-rel}} := \left\{ \begin{array}{l} ((\mathbf{F}, \mathbf{Y}), (\mathbf{W}, \mathbf{s})): \\ \mathbf{F} \in \mathcal{R}_q^{n \times m}; \mathbf{Y} \in \mathcal{R}_q^{n \times r}; \mathbf{W} \in \mathcal{R}^{m \times r} \\ \mathbf{s} \in \mathcal{R}^r; \|s_i\|_{\text{op}} \leq \mu \wedge s_i \notin 2\mathcal{R} \quad \forall i \in [r] \\ \|\mathbf{W}\| \leq \beta \wedge \mathbf{F}\mathbf{W} = \mathbf{Y} \text{diag}(\mathbf{s}) \bmod q \end{array} \right\}$$

Definition 21 (Relaxed Relation with Binary Field Claims). Let $m, n, \underline{n}, r \in \mathbb{N}$, $\beta, \mu \in \mathbb{R}$. We define the relation

$$\Xi_{\mathcal{R}, q, m, n, \underline{n}, r, \beta, \mu}^{\text{bin-lin-rel}} := \left\{ \begin{array}{l} ((\mathbf{F}, \mathbf{Y}, \mathbf{B}, \mathbf{u}), (\mathbf{W}, \mathbf{s})): \\ ((\mathbf{F}, \mathbf{Y}), (\mathbf{W}, \mathbf{s})) \in \Xi_{\mathcal{R}, q, m, n, r, \beta, \mu}^{\text{lin-rel}} \\ \mathbf{B} \in \mathbb{F}^{\underline{n} \times m}; \mathbf{u}^\top \in \mathbb{F}^{\underline{n} \times r} \\ \mathbf{B}(\mathbf{W} \bmod 2) = \mathbf{u}^\top \text{diag}(\mathbf{s} \bmod 2) \end{array} \right\}$$

Definition 22 (Relaxed Binary Evaluation Relation). Let $m, n, r \in \mathbb{N}$, $\beta, \mu \in \mathbb{R}$. We assume all dimensions are powers of two. Let $\nu = \log_2(m \cdot r)$. We define the relation:

$$\Xi_{\mathcal{R}, q, m, n, r, \beta, \mu}^{\text{bin-eval-rel}} := \left\{ \begin{array}{l} ((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), (\mathbf{W}, \mathbf{s})): \\ ((\mathbf{F}, \mathbf{Y}), (\mathbf{W}, \mathbf{s})) \in \Xi_{\mathcal{R}, q, m, n, r, \beta, \mu}^{\text{lin-rel}} \\ \tilde{\mathbf{w}} = \text{MLE}[\text{vec}((\mathbf{W} \bmod 2) \text{diag}(\mathbf{s} \bmod 2)^{-1})] \\ \mathbf{r} \in \mathbb{F}^\nu, t \in \mathbb{F} \\ \tilde{\mathbf{w}}(\mathbf{r}) = t \end{array} \right\}$$

Definition 23 (Relaxed Binary Evaluation Relation with Field-Switch). Let $m, n, r \in \mathbb{N}$, $\beta, \mu \in \mathbb{R}$, and let $\phi: \mathbb{T} \rightarrow V \subseteq \mathbb{F}$ be a field-switch packing. We assume all dimensions are powers of two. Let $\nu = \log_2(m \cdot r)$. We define the relation:

$$\Xi_{\mathcal{R}, q, m, n, r, \beta, \mu, \phi}^{\text{bin-eval-sw-rel}} := \left\{ \begin{array}{l} ((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), (\mathbf{W}, \mathbf{s})): \\ ((\mathbf{F}, \mathbf{Y}), (\mathbf{W}, \mathbf{s})) \in \Xi_{\mathcal{R}, q, m, n, r, \beta, \mu}^{\text{lin-rel}} \\ \mathbf{w}' := \text{vec}((\mathbf{W} \bmod 2) \text{diag}(\mathbf{s} \bmod 2)^{-1}); P_V(\mathbf{w}') \in V^{mr} \\ \mathbf{r} \in \mathbb{T}^\nu, t \in \mathbb{T} \\ \text{MLE}[\phi^{-1}(P_V(\mathbf{w}'))](\mathbf{r}) = t \end{array} \right\}$$

3.2 Field Switch

Lemma 8 (Field Switch). Let $\mathcal{R}$ be a cyclotomic ring with conductor $\mathfrak{f}$ for which 2 is inert, and let $q \geq 2$. Let $m, n, r \in \mathbb{N}$ be powers of two, write $\nu := \log_2(mr)$, let $\beta, \beta', \mu \in \mathbb{R}$, and let $\phi: \mathbb{T} \rightarrow V \subseteq \mathbb{F}$ be a field-switch packing for a field $\mathbb{T}$ of degree 2^ι . Protocol Π^{sw} (Figure 2) is correct for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw}} \rightarrow \Xi_{\text{par}_{\text{out}}}^{\text{bin-eval}}$ with correctness error $\delta = (\nu + \iota)/|\mathbb{F}|$ and knowledge sound for $\Xi_{\text{par}'_{\text{in}}}^{\text{bin-eval-sw-rel}} \cup \Xi_{\text{par}_{\text{sis}}}^{\text{SIS}} \leftarrow \Xi_{\text{par}'_{\text{out}}}^{\text{bin-eval-rel}}$ with knowledge error $\kappa = (2\nu + \iota)/|\mathbb{F}|$, where

$$\begin{aligned} \text{par}_{\text{in}} &= (\mathcal{R}, q, m, n, r, \beta, \phi), & \text{par}_{\text{out}} &= (\mathcal{R}, q, m, n, r, \beta), \\ \text{par}'_{\text{in}} &= (\mathcal{R}, q, m, n, r, \beta', \mu, \phi), & \text{par}'_{\text{out}} &= (\mathcal{R}, q, m, n, r, \beta', \mu), \\ \text{par}_{\text{sis}} &= (\mathcal{R}, q, m, n, 2\mu\beta'). \end{aligned}$$

Proof. Correctness. Let $((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), \mathbf{W}) \in \Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw}}$, so that $\mathbf{F}\mathbf{W} = \mathbf{Y} \bmod q$, $\|\mathbf{W}\| \leq \beta$, and $\mathbf{w}' := \text{vec}(\mathbf{W} \bmod 2) \in V^{mr}$ with $\text{MLE}[\phi^{-1}(\mathbf{w}')](\mathbf{r}) = t$. The honest prover sends $t = \mathbf{E}(\mathbf{r}) \phi^{-1}(\mathbf{w}')$, and the first check passes as

$$\boldsymbol{\theta}^\top t = \boldsymbol{\theta}^\top \mathbf{E}(\mathbf{r}) \phi^{-1}(\mathbf{w}') = \tilde{\mathbf{eq}}(\mathbf{r})^\top \phi^{-1}(\mathbf{w}') = t.$$

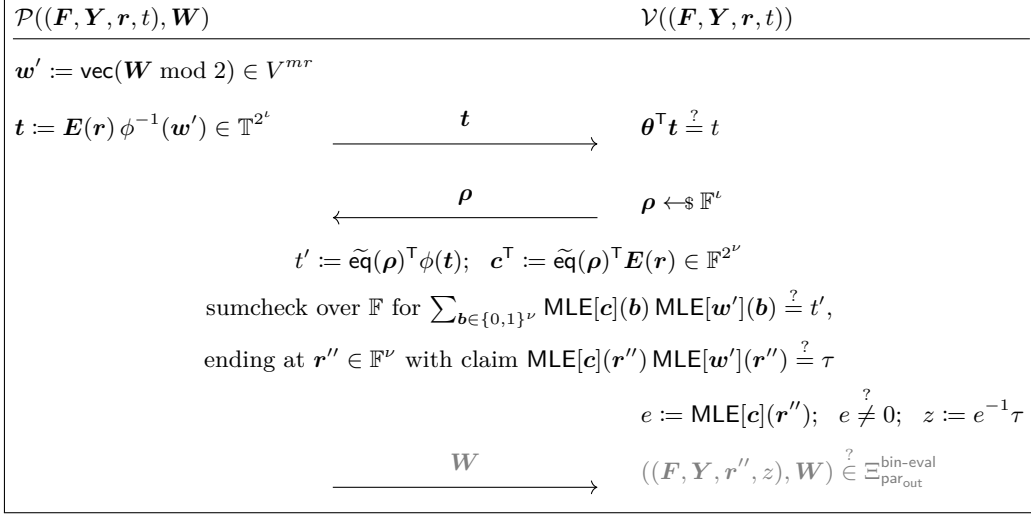


Figure 2: Protocol Π^{sw} , a reduction from $\Xi_{\text{par-in}}^{\text{bin-eval-sw}}$ to $\Xi_{\text{par-out}}^{\text{bin-eval}}$. Π^{sw} sends the marked parts only as a *proof* (but not *reduction*) of knowledge.

Commuting ϕ past $E(r)$ gives $\phi(t) = E(r) w'$, hence

$$t' = \tilde{eq}(\rho)^\top \phi(t) = \tilde{eq}(\rho)^\top E(r) w' = c^\top w' = \sum_{b \in \{0,1\}^\nu} \text{MLE}[c](b) \text{MLE}[w'](b),$$

so the sumcheck claim is true, the honest prover passes its ν rounds, and the final claim holds with $\tau = \text{MLE}[c](r'') \text{MLE}[w'](r'')$. Whenever $e \neq 0$ the verifier outputs $z = e^{-1} \tau = \text{MLE}[w'](r'')$, and $((F, Y, r'', z), W) \in \Xi_{\text{par-out}}^{\text{bin-eval}}$ (Definition 22), the lin part and the norm bound being untouched. It remains to bound the abort $e = 0$. By Lemma 4, $e = \tilde{eq}(\rho)^\top E(r) \tilde{eq}(r'') = \text{MLE}[\text{vec}(E(r)^\top)]((r'', \rho))$. The matrix $E(r)$ is nonzero, as $\theta^\top E(r) \mathbf{1} = \tilde{eq}(r)^\top \mathbf{1} = 1$, so $\text{MLE}[\text{vec}(E(r)^\top)]$ is a nonzero multilinear polynomial in $\nu + \iota$ variables and vanishes at the uniform point (r'', ρ) with probability at most $(\nu + \iota)/|\mathbb{F}|$ by the Schwartz–Zippel lemma.

Knowledge soundness. Without loss of generality, suppose $\mathcal{P}^*$ is deterministic and let ϵ be the probability, over the verifier randomness, that $\langle \mathcal{P}^*, \mathcal{V} \rangle((F, Y, r, t))$ outputs an instance–witness pair in $\Xi_{\text{par-out}}^{\text{bin-eval-rel}}$. $\mathcal{E}$ runs $\mathcal{P}^*$ once on fresh verifier randomness. If the output $((F, Y, r'', z), (W, s))$ does not lie in $\Xi_{\text{par-out}}^{\text{bin-eval-rel}}$, $\mathcal{E}$ aborts. Otherwise $((F, Y), (W, s)) \in \Xi^{\text{lin-rel}}$, $\|W\| \leq \beta'$, and $\text{MLE}[w'](r'') = z$ for $w' := \text{vec}((W \bmod 2) \text{diag}(s \bmod 2)^{-1})$, and $\mathcal{E}$ rewinds $\mathcal{P}^*$ and re-runs it on fresh verifier randomness until a second output in $\Xi_{\text{par-out}}^{\text{bin-eval-rel}}$, with witness $(\hat{W}, \hat{s})$ and $\hat{w}'$ defined accordingly.

We bind the two witnesses to F . Both satisfy their $\Xi^{\text{lin-rel}}$ claims, $FW = Y \text{diag}(s) \bmod q$ and $F\hat{W} = Y \text{diag}(\hat{s}) \bmod q$, so $F(W \text{diag}(\hat{s}) - \hat{W} \text{diag}(s)) = 0 \bmod q$. If $W \text{diag}(\hat{s}) - \hat{W} \text{diag}(s) \neq 0$, any nonzero column of it is a solution to $\Xi_{\text{par-sis}}^{\text{SIS}}$, of norm at most $2\mu\beta'$ as $\|s_j\|_{\text{op}}, \|\hat{s}_j\|_{\text{op}} \leq \mu$, and $\mathcal{E}$ outputs it and terminates. Otherwise the equality $W \text{diag}(\hat{s}) = \hat{W} \text{diag}(s)$ holds over $\mathcal{R}$. Reducing it modulo 2 and dividing by the invertible $\text{diag}(s \bmod 2) \text{diag}(\hat{s} \bmod 2)$, invertibility holding as $s_j, \hat{s}_j \notin 2\mathcal{R}$, gives $w' = \hat{w}'$, and $\mathcal{E}$ outputs (W, s) as the candidate witness for the input instance (F, Y, r, t) .

The conditions of the candidate in $\Xi_{\text{par-in}}^{\text{bin-eval-sw-rel}}$ other than the switch claim

$$\text{MLE}[\phi^{-1}(P_V(w'))](r) = t$$

are shared with $\Xi_{\text{par-out}}^{\text{bin-eval-rel}}$. Call a vector $w' \in \mathbb{F}^{2^\nu}$ *bad* if it violates the switch claim. $\mathcal{E}$

fails despite a successful first run exactly when the $\mathbf{w}'$ of that run is bad and the second run returns $\hat{\mathbf{w}}' = \mathbf{w}'$. We first bound, for a fixed bad $\mathbf{w}'$, the probability over the verifier randomness of a single run that the run outputs a witness in $\Xi_{\text{par}_{\text{out}}}^{\text{bin-eval-rel}}$ whose vector $\hat{\mathbf{w}}'$ equals $\mathbf{w}'$.

Fix a bad $\mathbf{w}'$. The message $\mathbf{t}$ precedes every challenge, so it is the same in every run as $\mathcal{P}^*$ is deterministic. Set $g := \text{MLE}[\mathbf{c}] \cdot \text{MLE}[\mathbf{w}']$ and $S := \mathbf{c}^\top \mathbf{w}' = \sum_{\mathbf{b} \in \{0,1\}^\nu} g(\mathbf{b})$, with $\mathbf{c}$ determined by the challenge $\boldsymbol{\rho}$ of the run. If the run outputs a witness in $\Xi_{\text{par}_{\text{out}}}^{\text{bin-eval-rel}}$ with $\hat{\mathbf{w}}' = \mathbf{w}'$, then $e \neq 0$ and $\tau = ez = \text{MLE}[\mathbf{c}](\mathbf{r}'') \text{MLE}[\mathbf{w}'](\mathbf{r}'') = g(\mathbf{r}'')$. By the soundness of the sumcheck protocol [LFKN92] for round polynomials of degree at most 2, if $t' \neq S$ then the event that every round check passes and the final claim satisfies $\tau = g(\mathbf{r}'')$ has probability at most $2\nu/|\mathbb{F}|$ over the round challenges, which are sampled after g is fixed. The equality $t' = S$ reads $\tilde{\mathbf{eq}}(\boldsymbol{\rho})^\top (\phi(\mathbf{t}) - \mathbf{E}(\mathbf{r}) \mathbf{w}') = 0$. The vector $\phi(\mathbf{t}) - \mathbf{E}(\mathbf{r}) \mathbf{w}'$ is fixed before $\boldsymbol{\rho}$ is sampled. If it is nonzero, its multilinear extension is a nonzero multilinear polynomial in ι variables, which vanishes at the uniform $\boldsymbol{\rho}$ with probability at most $\iota/|\mathbb{F}|$. If it is zero, then

$$\mathbf{E}(\mathbf{r}) \mathbf{w}' = \phi(\mathbf{t}).$$

The right side lies in $V^{2'}$, so applying P_V , which fixes it and commutes with $\mathbf{E}(\mathbf{r})$, gives $\mathbf{E}(\mathbf{r}) P_V(\mathbf{w}') = \phi(\mathbf{t})$. Applying ϕ^{-1} and the accepting first check $\boldsymbol{\theta}^\top \mathbf{t} = t$,

$$t = \boldsymbol{\theta}^\top \mathbf{t} = \boldsymbol{\theta}^\top \mathbf{E}(\mathbf{r}) \phi^{-1}(P_V(\mathbf{w}')) = \tilde{\mathbf{eq}}(\mathbf{r})^\top \phi^{-1}(P_V(\mathbf{w}')) = \text{MLE}[\phi^{-1}(P_V(\mathbf{w}'))](\mathbf{r}),$$

so $\mathbf{w}'$ satisfies the switch claim, contradicting that it is bad. Hence a single run outputs a witness in $\Xi_{\text{par}_{\text{out}}}^{\text{bin-eval-rel}}$ with $\hat{\mathbf{w}}' = \mathbf{w}'$ with probability at most $(2\nu + \iota)/|\mathbb{F}| = \kappa$.

The second run of $\mathcal{E}$ is a fresh run conditioned on outputting a witness in $\Xi_{\text{par}_{\text{out}}}^{\text{bin-eval-rel}}$, an event of probability ϵ , so for a fixed bad $\mathbf{w}'$ it returns $\hat{\mathbf{w}}' = \mathbf{w}'$ with probability at most κ/ϵ . The first run succeeds with a bad $\mathbf{w}'$ with probability at most ϵ , so $\mathcal{E}$ fails with probability at most $\epsilon \cdot \kappa/\epsilon = \kappa$. In every other case $\mathcal{E}$ outputs a SIS solution or a candidate $(\mathbf{W}, \mathbf{s})$ whose $\mathbf{w}'$ satisfies the switch claim, in which case $((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), (\mathbf{W}, \mathbf{s})) \in \Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw-rel}}$. Therefore $\mathcal{E}$ outputs a witness for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw-rel}} \cup \Xi_{\text{par}_{\text{sis}}}^{\text{SIS}}$ with probability at least $\epsilon - \kappa$.

Expected runtime of $\mathcal{E}$. $\mathcal{E}$ invokes $\mathcal{P}^*$ once and aborts on failure, and on success rewinds with fresh verifier randomness until a second success. The number of invocations satisfies $\mathbb{E}[T] \leq (1 - \epsilon) + \epsilon(1 + 1/\epsilon) = 2$, so $\mathcal{E}$ runs in expected polynomial time. $\square$

3.3 Left-expansion

$\mathcal{P}((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), \mathbf{W})$	$\mathcal{V}((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t))$
$\mathbf{r}^\top := (\mathbf{r}_0^\top \ \mathbf{r}_1^\top) \in \mathbb{F}^{\log_2(m)} \times \mathbb{F}^{\log_2(r)}$	
$\mathbf{u}^\top := \tilde{\mathbf{eq}}(\mathbf{r}_0)^\top \mathbf{W} \bmod 2 \in \mathbb{F}^r$	$\mathbf{u}^\top \tilde{\mathbf{eq}}(\mathbf{r}_1) \stackrel{?}{=} t$
$\mathbf{B} := \tilde{\mathbf{eq}}(\mathbf{r}_0)^\top \in \mathbb{F}^{1 \times m}$	$((\mathbf{F}, \mathbf{Y}, \mathbf{B}, \mathbf{u}), \mathbf{W}) \stackrel{?}{\in} \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$

Figure 3: Protocol $\Pi^{\text{left-exp}}$, a reduction from $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval}}$ to $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$. $\Pi^{\text{left-exp}}$ sends the marked parts only as a *proof* (but not *reduction*) of knowledge.

Lemma 9 (Left-expansion). *Let $\mathcal{R}$ be a cyclotomic ring with conductor $\mathfrak{f}$ for which 2 is inert, and let $q \geq 2$. Let $m, n, r \in \mathbb{N}$ be powers of two, write $\nu := \log_2(mr)$, and let*

$\beta, \beta', \mu \in \mathbb{R}$. Protocol $\Pi^{\text{left-exp}}$ (Figure 3) is correct for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval}} \rightarrow \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$ and knowledge sound for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-rel}} \leftarrow \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin-rel}}$ with knowledge error $\kappa = 0$, where

$$\begin{aligned} \text{par}_{\text{in}} &= (\mathcal{R}, q, m, n, r, \beta), & \text{par}_{\text{out}} &= (\mathcal{R}, q, m, n, 1, r, \beta), \\ \text{par}'_{\text{in}} &= (\mathcal{R}, q, m, n, r, \beta', \mu), & \text{par}'_{\text{out}} &= (\mathcal{R}, q, m, n, 1, r, \beta', \mu). \end{aligned}$$

Proof. Correctness. Let $((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), \mathbf{W}) \in \Xi_{\text{par}_{\text{in}}}^{\text{bin-eval}}$, so that $\mathbf{F}\mathbf{W} = \mathbf{Y} \bmod q$, $\|\mathbf{W}\| \leq \beta$, and $\tilde{\mathbf{w}}(\mathbf{r}) = t$ for $\tilde{\mathbf{w}} := \text{MLE}[\text{vec}(\mathbf{W} \bmod 2)]$. The honest prover sends $\mathbf{u}^\top = \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_0)^\top \mathbf{W} \bmod 2 \in \mathbb{F}^r$. By Lemma 4 over $\mathbb{F}$,

$$\mathbf{u}^\top \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_1) = \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_0)^\top (\mathbf{W} \bmod 2) \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_1) = \tilde{\mathbf{w}}(\mathbf{r}) = t,$$

so the check succeeds. The binary claim of $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$ (Definition 21) with $\mathbf{B} = \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_0)^\top$ reads $\mathbf{B}\mathbf{W} = \mathbf{u}^\top \bmod 2$, which holds by definition of $\mathbf{u}$, while the lin part $((\mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{lin}}$ is unchanged. As $\|\mathbf{W}\| \leq \beta$, we conclude $((\mathbf{F}, \mathbf{Y}, \mathbf{B}, \mathbf{u}), \mathbf{W}) \in \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$.

Knowledge soundness. Without loss of generality, suppose $\mathcal{P}^*$ is deterministic. The verifier is deterministic as well, so $\langle \mathcal{P}^*, \mathcal{V} \rangle((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t))$ is a single fixed run. $\mathcal{E}$ executes it once, invoking $\mathcal{P}^*$ once, and outputs the witness $(\mathbf{W}, \mathbf{s})$ of its output $((\mathbf{F}, \mathbf{Y}, \mathbf{B}, \mathbf{u}), (\mathbf{W}, \mathbf{s}))$ as the candidate for the input instance $((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t))$. It remains to show that whenever the output lies in $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin-rel}}$, the candidate lies in $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-rel}}$, so that $\kappa = 0$.

By Definition 21 we have $((\mathbf{F}, \mathbf{Y}), (\mathbf{W}, \mathbf{s})) \in \Xi_{\text{par}'_{\text{in}}}^{\text{lin-rel}}$, with $\|\mathbf{W}\| \leq \beta'$, and the binary claim $\mathbf{B}\mathbf{W} = \mathbf{u}^\top \text{diag}(\mathbf{s}) \bmod 2$, that is $\tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_0)^\top (\mathbf{W}_{:,j} \bmod 2) = u_j (s_j \bmod 2)$ for every column $j \in [r]$. Since $s_j \notin 2\mathcal{R}$ and $\mathbb{F}$ is a field, each $s_j \bmod 2$ is invertible, so dividing column j by it and summing against $\tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_1)$, Lemma 4 over $\mathbb{F}$ gives

$$\tilde{\mathbf{w}}(\mathbf{r}) = \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_0)^\top (\mathbf{W} \bmod 2) \text{diag}(\mathbf{s} \bmod 2)^{-1} \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_1) = \sum_{j \in [r]} u_j \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_1)_j = t,$$

for $\tilde{\mathbf{w}} = \text{MLE}[\text{vec}((\mathbf{W} \bmod 2) \text{diag}(\mathbf{s} \bmod 2)^{-1})]$, the last equality being the accepting check. Together with $\|\mathbf{W}\| \leq \beta'$, this places $((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), (\mathbf{W}, \mathbf{s})) \in \Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-rel}}$.

Expected runtime of $\mathcal{E}$. $\mathcal{E}$ invokes $\mathcal{P}^*$ once, so $\mathcal{E}$ runs in expected polynomial time. $\square$

3.4 Folding

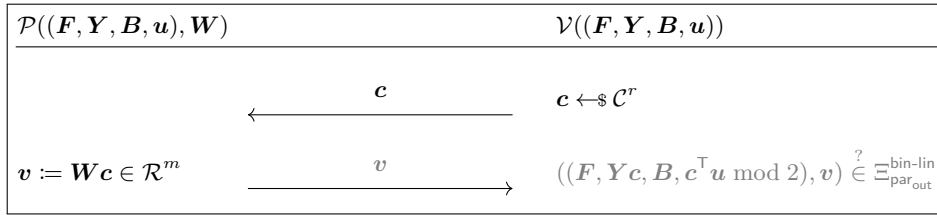


Figure 4: Protocol Π^{fold} , a reduction from $\Xi_{\text{par}_{\text{in}}}^{\text{bin-lin}}$ to $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$. Π^{fold} sends the marked parts only as a *proof* (but not *reduction*) of knowledge.

Lemma 10 (Folding). *Let $\mathcal{R}$ be a cyclotomic ring with conductor $\mathfrak{f}$ for which 2 is inert and let $\mathcal{C} \subseteq \mathcal{R}$ be a challenge set that is injective modulo 2, with expansion factor $\gamma_{\mathcal{C}}$. Let $q, m, n, \underline{n}, r \in \mathbb{N}$ and $\beta, \beta' \in \mathbb{R}$. Protocol Π^{fold} (Figure 4) is correct for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-lin}} \rightarrow \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$ and knowledge sound for $\Xi_{\text{par}'_{\text{in}}}^{\text{bin-lin-rel}} \leftarrow \Xi_{\text{par}'_{\text{out}}}^{\text{bin-lin}}$ with knowledge error $\kappa = r/|\mathcal{C}|$, where*

$$\begin{aligned} \text{par}_{\text{in}} &= (\mathcal{R}, q, m, n, \underline{n}, r, \beta), & \text{par}_{\text{out}} &= (\mathcal{R}, q, m, n, \underline{n}, 1, r \gamma_{\mathcal{C}} \beta), \\ \text{par}'_{\text{in}} &= (\mathcal{R}, q, m, n, \underline{n}, r, 2\beta', 2\gamma_{\mathcal{C}}), & \text{par}'_{\text{out}} &= (\mathcal{R}, q, m, n, \underline{n}, 1, \beta'). \end{aligned}$$

Proof. Correctness. Let $((F, Y, B, u), W) \in \Xi_{\text{par}_{\text{in}}}^{\text{bin-lin}}$, so $FW = Y \bmod q$, $BW = u^\top \bmod 2$ and $\|W\| \leq \beta$. The honest prover sends $v = Wc$. By linearity $Fv = Yc \bmod q$ and $Bv = u^\top c \bmod 2$, and

$$\begin{aligned} \|v\| &= \left\| \sum_{j \in [r]} W_{:,j} c_j \right\| \leq \sum_{j \in [r]} \|c_j\|_{\text{op}} \|W_{:,j}\| \leq \gamma_C \sum_{j \in [r]} \|W_{:,j}\| \\ &\leq r \gamma_C \|W\| \leq r \gamma_C \beta, \end{aligned}$$

where the first step is the triangle inequality with submultiplicativity of the operator norm, the second uses $\|c_j\|_{\text{op}} \leq \gamma_C$, and the third bounds each column norm by the columnwise maximum $\|W\|$. Hence $((F, Yc, B, c^\top u), v) \in \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$.

Knowledge soundness. Without loss of generality, suppose $\mathcal{P}^*$ is deterministic and let p be the probability, over the challenge $c \leftarrow \mathcal{C}^r$, that $\langle \mathcal{P}^*, \mathcal{V} \rangle((F, Y, B, u))$ outputs an instance–witness pair in $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$. $\mathcal{E}$ invokes the coordinate-wise special-soundness extractor of [FMN24, Lemma 7.1] on $\mathcal{P}^*$, which produces, with probability at least $p - r/|\mathcal{C}|$, accepting transcripts for challenges $c^{(0)}, \dots, c^{(r)} \in \mathcal{C}^r$ such that $c^{(j)}$ and $c^{(r)}$ differ only in coordinate j , together with their output witnesses $v^{(j)} \in \mathcal{R}^m$ in $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$, so that for $j \in \{0, \dots, r\}$

$$Fv^{(j)} = Yc^{(j)} \bmod q, \quad Bv^{(j)} = u^\top c^{(j)} \bmod 2, \quad \|v^{(j)}\| \leq \beta'.$$

For $j \in [r]$, subtracting the equations of transcripts j and r cancels every challenge coordinate except the j -th, so the tuple

$$w^{(j)} := v^{(j)} - v^{(r)}, \quad s^{(j)} := c_j^{(j)} - c_j^{(r)}$$

satisfies

$$\begin{aligned} Fw^{(j)} &= s^{(j)} y_j \bmod q, \quad Bw^{(j)} = s^{(j)} u_j \bmod 2, \\ \|w^{(j)}\| &\leq 2\beta', \quad \|s^{(j)}\|_{\text{op}} \leq 2\gamma_C, \end{aligned}$$

where y_j and u_j are the j -th columns of Y and $u^\top$. As $c_j^{(j)} \neq c_j^{(r)}$ and $\mathcal{C}$ is injective modulo 2, every $s^{(j)} \notin 2\mathcal{R}$, so the extractor's loss is $r/|\mathcal{C}| = \kappa$.

Assembling the columns $W := (w^{(j)})_{j \in [r]}$ and the slack $s := (s^{(j)})_{j \in [r]}$, the displayed equalities read $FW = Y \text{diag}(s) \bmod q$ and $BW = u^\top \text{diag}(s) \bmod 2$, with $\|W\| \leq 2\beta'$, $\|s^{(j)}\|_{\text{op}} \leq 2\gamma_C$ and $s^{(j)} \notin 2\mathcal{R}$. Hence, with probability at least $p - \kappa$, $\mathcal{E}$ outputs (W, s) with $((F, Y, B, u), (W, s)) \in \Xi_{\text{par}_{\text{in}}}^{\text{bin-lin-rel}}$.

Expected runtime of $\mathcal{E}$. By Lemma 3, $\mathcal{E}$ invokes $\mathcal{P}^*$ at most $r + 1$ times in expectation, so it runs in expected polynomial time. $\square$

3.5 Composition

Corollary 1 (Composition). *Let $\mathcal{R}$ be a cyclotomic ring with conductor $\mathfrak{f}$ for which 2 is inert, let $q \geq 2$, let $m, n, r \in \mathbb{N}$ be powers of two with $\nu := \log_2(mr)$, let $\phi: \mathbb{T} \rightarrow V \subseteq \mathbb{F}$ be a field-switch packing, and let $\mathcal{C} \subseteq \mathcal{R}$ be injective modulo 2 with expansion factor γ_C . Let $\beta, \beta' \in \mathbb{R}$. The composition Π of Π^{sw} , $\Pi^{\text{left-exp}}$, and Π^{fold} is correct for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw}} \rightarrow \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$ with correctness error $\delta = (\nu + \iota)/|\mathbb{F}|$ and knowledge sound for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw-rel}} \cup \Xi_{\text{par}_{\text{sis}}}^{\text{SIS}} \leftarrow \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$ with knowledge error $\kappa = (2\nu + \iota)/|\mathbb{F}| + r/|\mathcal{C}|$, where*

$$\begin{aligned} \text{par}_{\text{in}} &= (\mathcal{R}, q, m, n, r, \beta, \phi), & \text{par}_{\text{out}} &= (\mathcal{R}, q, m, n, 1, 1, r \gamma_C \beta), \\ \text{par}'_{\text{in}} &= (\mathcal{R}, q, m, n, r, 2\beta', 2\gamma_C, \phi), & \text{par}'_{\text{out}} &= (\mathcal{R}, q, m, n, 1, 1, \beta'), \\ \text{par}_{\text{sis}} &= (\mathcal{R}, q, m, n, 8\gamma_C \beta'). \end{aligned}$$

Proof. Errors add under sequential composition [KLNO24], and the parameters chain through Lemmas 8 to 10 with $\underline{n} = 1$, $\mu = 2\gamma_C$, and norm $2\beta'$ from the extractor of Π^{fold} , so that the SIS bound $2\mu \cdot 2\beta'$ of Lemma 8 reads $8\gamma_C\beta'$. $\square$

Definition 24 (Packed Ajtai Commitment). Let $m, n, r \in \mathbb{N}$ be powers of two with $\nu := \log_2(mr)$, let $\beta, \mu \in \mathbb{R}$, and let $\phi: \mathbb{T} \rightarrow V \subseteq \mathbb{F}$ be a field-switch packing. The *packed Ajtai commitment* with parameters $(m, n, r, \beta, \mu, \phi)$ is the commitment scheme over $\mathbb{T}^{2^\nu}$ given by:

- **Setup**(1^λ) samples and outputs $\mathbf{F} \leftarrow \mathcal{R}_q^{n \times m}$.
- **Com**($\mathbf{F}, \mathbf{f}$) computes $\mathbf{W} \in \mathcal{R}^{m \times r}$ with $\text{vec}(\mathbf{W}) = \text{lift}(\phi(\mathbf{f}))$ and returns the commitment $\mathbf{Y} := \mathbf{F}\mathbf{W} \bmod q$ with opening state $(\mathbf{W}, \mathbf{1})$.
- **Open**($\mathbf{F}, \mathbf{Y}, \mathbf{f}, (\mathbf{W}, \mathbf{s})$) checks $\|\mathbf{W}\| \leq \beta$, $\|s_j\|_{\text{op}} \leq \mu$ and $s_j \notin 2\mathcal{R}$ for all $j \in [r]$, $\mathbf{F}\mathbf{W} = \mathbf{Y} \text{diag}(\mathbf{s}) \bmod q$, and $\phi^{-1}(P_V(\text{vec}((\mathbf{W} \bmod 2) \text{diag}(\mathbf{s} \bmod 2)^{-1}))) = \mathbf{f}$.

Theorem 1 (Polynomial Commitment Scheme over $\mathbb{T}$). *In the setting of Corollary 1, let $\gamma_C \geq 1$, let $\beta \geq \sqrt{\varphi m}$, let $\beta' := r\gamma_C\beta$, and let $\text{SIS}_{\text{par}_{\text{sis}}}$ be hard with security parameter λ . Let Π be the protocol of Corollary 1 run as a proof of knowledge: the prover sends the final witness $\mathbf{v}$ and the verifier checks $((\mathbf{F}, \mathbf{Y}\mathbf{c}, \mathbf{B}, \mathbf{c}^\top \mathbf{u}), \mathbf{v}) \in \Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$. Then the packed Ajtai commitment with parameters $(m, n, r, 2\beta', 2\gamma_C, \phi)$ together with Π is a multilinear polynomial commitment scheme for ν variables over $\mathbb{T}$ with binding error $2^{-\lambda}$, correctness error δ and, against probabilistic polynomial-time provers, knowledge error $\kappa + 2^{-\lambda}$.*

Proof. For completeness, an honest $\mathbf{W}$ has binary coefficients, so $\|\mathbf{W}\| \leq \sqrt{\varphi m} \leq 2\beta'$, and $\text{vec}(\mathbf{W} \bmod 2) = \phi(\mathbf{f}) \in V^{2^\nu}$, so **Open** accepts the state $(\mathbf{W}, \mathbf{1})$, as $\|\mathbf{1}\|_{\text{op}} = 1 \leq 2\gamma_C$. For binding, two openings with $\mathbf{f} \neq \mathbf{f}'$ have distinct projections $P_V(\mathbf{w}') \neq P_V(\mathbf{w}'')$ of their vectors $\mathbf{w}' = \text{vec}((\mathbf{W} \bmod 2) \text{diag}(\mathbf{s} \bmod 2)^{-1})$ and $\mathbf{w}'' = \text{vec}((\mathbf{W}' \bmod 2) \text{diag}(\mathbf{s}' \bmod 2)^{-1})$, hence $\mathbf{w}' \neq \mathbf{w}''$, so the two matrices differ in some column j and $\mathbf{z} := s'_j \mathbf{W}_{:,j} - s_j \mathbf{W}'_{:,j}$ is nonzero, satisfies $\mathbf{F}\mathbf{z} = 0 \bmod q$, and has norm at most $2 \cdot 2\gamma_C \cdot 2\beta'$, a solution of $\text{SIS}_{\text{par}_{\text{sis}}}$, so a probabilistic polynomial-time adversary produces two such openings with probability at most $2^{-\lambda}$ by Definition 5. With $\mathbf{F}, \mathbf{Y}$, and $(\mathbf{W}, \mathbf{s})$ as public parameter, commitment, and opening state, the evaluation relation of the scheme is $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw-rel}}$, and an output $(\mathbf{Y}, (\mathbf{W}, \mathbf{1}))$ of **Com** with $\text{MLE}[\mathbf{f}](\mathbf{r}) = t$ gives $((\mathbf{F}, \mathbf{Y}, \mathbf{r}, t), \mathbf{W}) \in \Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw}}$, so by Corollary 1 Π is a proof of knowledge for Ξ^{eval} with errors δ and κ , up to the SIS branch of the extraction relation. Correctness holds only on these exact openings, i.e. with slack $\mathbf{1}$ recovering $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw}}$, and norm at most β , whereas the extractor returns relaxed openings $(\mathbf{W}, \mathbf{s})$ with $\|\mathbf{W}\| \leq 2\beta'$ and $\|s_j\|_{\text{op}} \leq 2\gamma_C$, i.e. for $\Xi_{\text{par}_{\text{in}}}^{\text{bin-eval-sw-rel}}$. **Open** accepts the relaxed opening, and binding as shown above covers all accepted openings, so the relaxed extraction is sufficient for Definition 19. With a probabilistic polynomial-time prover, the extractor is an expected polynomial-time algorithm on $\mathbf{F} \leftarrow \mathcal{R}_q^{n \times m}$, so this branch has probability at most $2^{-\lambda}$ by Definition 5. $\square$

Lemma 11 (Efficiency). *In the protocol Π of Theorem 1, with $2^\nu = mr$, the prover performs $O(2^\nu)$ multiplications in $\mathbb{T}$ and in $\mathbb{F}$, $O(2^{\nu+\iota})$ additions in $\mathbb{T}$ and in $\mathbb{F}$, and $O(2^\nu)$ multiplications of a ring element by a challenge in $\mathcal{C}$. The verifier performs $O(2^\iota)$ multiplications in $\mathbb{T}$, $O(\nu 2^{2\iota} + m + r)$ multiplications in $\mathbb{F}$, $O(nm)$ multiplications in $\mathcal{R}_q$, and $O(nr)$ multiplications by challenges in $\mathcal{C}$. The prover sends 2^ι elements of $\mathbb{T}$, $3\nu + r$ elements of $\mathbb{F}^2$, and m elements of $\mathcal{R}$ of norm at most β' , and the verifier sends $\nu + \iota$ elements of $\mathbb{F}$ and r elements of $\mathcal{C}$.*

²The 3ν sumcheck elements can be reduced to 2ν : the verifier knows $g_i(0) + g_i(1)$ from the previous round, so two further values of the degree-2 round polynomial g_i determine it.

Proof. In Π^{sw} , the prover computes $\tilde{\text{eq}}(\mathbf{r})$ with 2^ν multiplications in $\mathbb{T}$, then $\mathbf{t}$ by adding $f_{\mathbf{b}}$ into $t_{\mathbf{k}}$ for every pair with $\mathbf{E}(\mathbf{r})_{\mathbf{k},\mathbf{b}} = 1$, at most $2^{\nu+\iota}$ additions in $\mathbb{T}$, and $\mathbf{c}$ likewise from $\tilde{\text{eq}}(\boldsymbol{\rho})$ with at most $2^{\nu+\iota}$ additions in $\mathbb{F}$. The sumcheck for the degree-2 product $\text{MLE}[\mathbf{c}] \cdot \text{MLE}[\mathbf{w}']$ costs the prover $O(2^\nu)$ multiplications in $\mathbb{F}$, halving the tables in every round, and sends three elements of $\mathbb{F}$ per round. The verifier checks $\boldsymbol{\theta}^\top \mathbf{t} = t$ and computes t' with 2^ι multiplications each, runs the ν sumcheck rounds with $O(1)$ multiplications per round, and computes $e = \tilde{\text{eq}}(\boldsymbol{\rho})^\top \mathbf{E}(\mathbf{r}) \tilde{\text{eq}}(\mathbf{r}'')$ without materialising $\mathbf{c}$: the vector $\mathbf{E}(\mathbf{r}) \tilde{\text{eq}}(\mathbf{r}'') \in \mathbb{F}^{2^\iota}$ holds the coordinates over the basis $(\theta_{\mathbf{k}} \otimes 1)_{\mathbf{k}}$ of

$$\sum_{\mathbf{b} \in \{0,1\}^\nu} \text{eq}(\mathbf{r}, \mathbf{b}) \otimes \text{eq}(\mathbf{r}'', \mathbf{b}) = \prod_{i \in [\nu]} ((1 - r_i) \otimes (1 - r''_i) + r_i \otimes r''_i) \in \mathbb{T} \otimes_{\mathbb{F}_2} \mathbb{F},$$

which are ν multiplications in the $\mathbb{F}$ -algebra $\mathbb{T} \otimes_{\mathbb{F}_2} \mathbb{F}$ of dimension 2^ι , each $O(2^{2^\iota})$ multiplications in $\mathbb{F}$, followed by the inner product with $\tilde{\text{eq}}(\boldsymbol{\rho})$. In $\Pi^{\text{left-exp}}$, the prover computes $\tilde{\text{eq}}(\mathbf{r}_0)$ and $\mathbf{u}^\top = \tilde{\text{eq}}(\mathbf{r}_0)^\top \mathbf{W} \bmod 2$ with $O(mr)$ multiplications in $\mathbb{F}$, and the verifier checks $\mathbf{u}^\top \tilde{\text{eq}}(\mathbf{r}_1) = t$ with $O(r)$. In Π^{fold} , the prover computes $\mathbf{v} = \mathbf{W}\mathbf{c}$ with mr multiplications by challenges, and the verifier checks $\mathbf{F}\mathbf{v} = \mathbf{Y}\mathbf{c} \bmod q$ with nm multiplications in $\mathcal{R}_q$ and nr multiplications by challenges, $\mathbf{B}\mathbf{v} = \mathbf{u}^\top \mathbf{c} \bmod 2$ with $O(m+r)$ multiplications in $\mathbb{F}$, and $\|\mathbf{v}\| \leq \beta'$ with $O(m\varphi)$ integer operations. The messages are $\mathbf{t} \in \mathbb{T}^{2^\iota}$, the 3ν sumcheck coefficients, $\mathbf{u} \in \mathbb{F}^r$, and $\mathbf{v} \in \mathcal{R}^m$, against the challenges $\boldsymbol{\rho} \in \mathbb{F}^\iota$, $\mathbf{r}'' \in \mathbb{F}^\nu$, and $\mathbf{c} \in \mathcal{C}^r$. $\square$

4 Recursion into LaBRADOR

At the end of Π^{fold} , the prover would ordinarily reveal the folded witness $\mathbf{v}$, commitment matrix $\mathbf{Y}$, and left expansion $\mathbf{u}$, which satisfy the folded equations and norm bound

$$\mathbf{F}\mathbf{v} = \mathbf{Y}\mathbf{c} \bmod q, \quad \mathbf{B}\mathbf{v} = \mathbf{u}^\top \mathbf{c} \bmod 2, \quad \|\mathbf{v}\| \leq \beta'.$$

Our recursive variant replaces this large final message with Ajtai commitments to $\mathbf{Y}$ and $\mathbf{u}$, followed by one LaBRADOR proof that opens them and proves the folded relation. The commitments are fixed before the folding challenge, so the prover cannot choose either object after seeing the coefficients $\mathbf{c}$.

The main technical obstacle is that the folded equations use $\mathcal{R}_q$ and $\mathcal{R}_2$, whereas LaBRADOR works over the smaller power-of-two ring $\mathcal{R}'_Q$ with its separate modulus Q . We bridge the rings by cutting every witness ring element into coefficient chunks and every public multiplier into blocks small enough that their products are computed in $\mathcal{R}'_Q$ without wraparound. The resulting block equations are connected by short carry ring elements, including two correction terms that implement reduction by $\Phi_{3^\kappa}(\mathbf{x})$. An RNS encoding handles the equations modulo q , while an integer lift followed by reduction modulo 2 handles the binary equation and the claimed multilinear evaluation. Quotients and carries are gadget-decomposed into bounded digits so that all auxiliary witnesses meet LaBRADOR's norm requirements. Random linear tests enforce that chunks have zero coefficients outside their prescribed supports, and an explicit no-wrap bound turns the identities proved modulo Q back into integer identities.

These checks form a standard LaBRADOR relation and give the completeness and knowledge-soundness guarantees of Theorem 2, and Appendix A presents the full details of this mechanical translation.

5 Concrete Instantiation

5.1 Implementation

We implement the commitment, the fold, and their verifier in Rust,³ with the arithmetic kernels written for AVX-512. The prover and verifier use the same transforms and inner-product routines, with the verifier applying them to the shorter folded opening. All kernels use one thread. The implementation supports clear openings, bit-dropped commitments, and recursive openings through LaBRADOR.

5.1.1 Binary Extension Field

For $f = 3^5 = 243$, the binary residue field is

$$\mathbb{F} = \mathcal{R}_2 \cong \mathbb{F}_2[x] / (x^{162} + x^{81} + 1).$$

An element of $\mathbb{F}$ has 162 binary coefficients and is stored in three machine words. Within each word, the bits are the coefficients of consecutive powers of x . For example, the bit string 10 represents x , and multiplying 10 by 10 gives 100, which represents x^2 . When several terms contribute to the same coefficient, they are combined modulo two by exclusive-or. The word-sliced SIMD layout places the first word of several field elements in one vector register, their second word in another register, and their third word in a third register. Write two elements as $a_0 + a_1x^{64} + a_2x^{128}$ and $b_0 + b_1x^{64} + b_2x^{128}$. We apply VPCLMULQDQ carry-less multiplication instruction to the three diagonal products $a_i b_i$ and the three cross products $(a_i + a_j)(b_i + b_j)$ in parallel across vector lanes. These six products determine all coefficients of the unreduced product. Reduction uses the identity $x^{162} = x^{81} + 1$ in $\mathbb{F}$ and folds the high coefficients back twice. An inner product can therefore accumulate the six unreduced product components for every summand without coefficient growth and perform this reduction only once at the end.

The source field $\mathbb{T} = \mathbb{F}_{2^{128}}$ is represented in the GHASH basis, with modulus $x^{128} + x^7 + x^2 + x + 1$. We use the $\mathbb{F}_2$ -linear packing map

$$\phi \left(\sum_{i=0}^{127} a_i x^i \right) := \sum_{i=0}^{127} a_i x^i \in \mathbb{F}.$$

Thus ϕ copies the GHASH coefficient vector into the first 128 power-basis coordinates of $\mathbb{F}$ and fills the remaining coordinates with zero. It preserves addition and multiplication by binary scalars, but it does not preserve multiplication of field elements. We instantiate the field switch of Section 2.7 with $\iota = 7$. The prover first sends the $2^7 = 128$ partial evaluations over $\mathbb{T}$. The verifier checks that their GHASH-basis recombination equals the claimed evaluation, maps them coefficientwise through ϕ , and samples a random linear combination over $\mathbb{F}$. The remaining variables are handled by a sumcheck over $\mathbb{F}$ using the word-sliced multiplication kernel above.

5.1.2 Ring Arithmetic and NTT Transformation

The committed ring elements are represented in the cyclotomic superring $\mathcal{S} = \mathbb{Z}[\zeta_{8f}]$. The embedding $\mathcal{R} \rightarrow \mathcal{S}$ maps ζ_f to $-\zeta_{8f}^4$. The superring is a free module of rank four over $\mathcal{R}$. We pack four consecutive elements of $\mathbb{F}$ into one element of $\mathcal{S}$ by interleaving their bits across the four module components. Multiplication by an element of $\mathcal{R}$, including every folding challenge, acts identically on these components. The commitment and fold can therefore process the packed elements without separating them. The analysis of Section 3 carries over verbatim: $\mathbf{W}$ lives in $\mathcal{S}^{m \times r}$ and $\mathbf{F}$ in $\mathcal{S}_q^{n \times m}$, the challenges and slacks stay

³The source code is available at <https://github.com/osdnk/labinius>.

in $\mathcal{R}$ and act $\mathcal{R}$ -linearly on the four components, and the binary claims are read on each component separately, as 2 ramifies in $\mathcal{S}$ and $\mathcal{S}/2\mathcal{S}$ is not a field. The SIS instances estimated below are accordingly over $\mathcal{S}$.

Each residue occupies one signed SIMD lane, and a vector register holds the same coefficient of several independent ring elements. The NTT follows the mixed-radix factorisation of the superring conductor. For fully splitting primes, the transform ends in linear slots. We also support primes for which the final factors are irreducible quadratics. Multiplication in these slots uses a Karatsuba formula, and the output is converted to the same public module layout as in the fully splitting case. The reduction schedule is selected for each modulus from the available signed-lane headroom. Smaller moduli admit lazy reduction across several levels. The supported primes are 3889, 9721, 17497, and 19441 for fully splitting transforms, and 2917, 4861, and 12637 for transforms ending in quadratic factors. We select two or three primes for the commitment modulus according to the target SIS security level.

The binary input permits a specialised forward transform. The upper levels and the following twiddle factors form a linear function of a small group of input bits, so we replace them with byte-table lookups. The remaining butterflies use signed Montgomery arithmetic. For fully splitting small primes, the final levels are hand-scheduled in assembly so that an entire transform block remains in vector registers. The verifier uses a generic transform with the same tree for the non-binary folded opening.

5.1.3 Commitment and Fold

The Ajtai commitment is one inner product per NTT slot and RNS limb. The public matrix $\mathbf{F}$ is stored in the transform’s centred slot layout and read once per commitment. Vector dot-product instructions accumulate signed lane products into wider lanes without reducing each product.

The commitment retains the transformed witness for the single RNS limb. The prover computes $\mathbf{v} = \mathbf{W}\mathbf{c}$ within this limb. Only the folded result is transformed back to its centred coefficient representation. The fold therefore reads the retained transform once and uses only the base limb. The verifier transforms $\mathbf{v}$ modulo every limb, recomputes $\mathbf{F}\mathbf{v}$ with the commitment kernel, and compares it with $\mathbf{Y}\mathbf{c}$. It also checks the coefficient norm of $\mathbf{v}$ and its evaluation claim.

The challenge set consists of signed elements of $\mathcal{R}$ with fixed Hamming weight, nonzero coefficients in $\{-1, 1\}$. We sample the support uniformly with a partial Fisher-Yates shuffle[FY38, Dur64]. The signs are deterministically derived by a keyed hash of the support. Since $-1 = 1 \bmod 2$, signing preserves the image of the challenge in $\mathbb{F}$ while reducing the average norm of sums of challenges. To control the operator norm, we evaluate the squared magnitude of the challenge in Fourier slots. The slot energy gives an upper bound on the operator norm.

5.1.4 Bit Dropping Optimisation

We reduce a clear commitment by omitting d low bits. The prover first applies the Garner algorithm to convert the RNS residues into mixed-radix digits. Coefficientwise, write $\mathbf{Y} = \mathbf{Y}_0 + 2^d \mathbf{Y}_1$, where $\mathbf{Y}_0$ is the rounding error. The prover sends $\mathbf{Y}_1$ together with the remaining mixed-radix digits. The verifier uses these digits to reconstruct the rounded commitment and computes $\mathbf{z}' := \mathbf{F}\mathbf{v} - 2^d \mathbf{Y}_1 \mathbf{c} = \mathbf{Y}_0 \mathbf{c} \bmod q$. It transforms only this residual back to coefficient form. The Garner algorithm then reconstructs the centred integer residual coefficientwise from its RNS limbs and accumulates its squared norm. The identity $(\mathbf{F} \ \mathbf{I})(\mathbf{v}, -\mathbf{z}') = 2^d \mathbf{Y}_1 \mathbf{c} \bmod q$ is a normal-form SIS relation. The residual $-\mathbf{z}'$ is the component corresponding to the identity block, and the extracted preimage is $(\mathbf{v}, -\mathbf{z}')$. For folded-witness bound B_v and residual bound $B_{z'}$, this preimage has bound $\sqrt{B_v^2 + B_{z'}^2}$.

5.1.5 Connection with Binius and Flock

We use the lattice commitment as the polynomial commitment layer beneath both Binius [Irr26] and Flock [BRW26]. For Binius, we retain its constraint reductions and packed binary witness, but replace its Merkle-tree commitment and BaseFold opening with our Ajtai commitment and folding protocol. These reductions produce an evaluation claim over $\mathbb{T}$, which the field-switch protocol converts into a claim over $\mathbb{F}$ for our commitment to open. For Flock, we likewise retain its R1CS zero-check and linear-check, bind the lattice commitment into its transcript, and pass the resulting evaluation claim through the same field switch. Both front ends end their reductions on the bits of the packed trace rather than on the words, with ν multilinear coordinates over the words and, within a word, ι multilinear coordinates for Binius and a univariate skip over six of them for Flock. Instead of first reducing this to a claim on the words over $\mathbb{T}$ by the ring switching of [DP25], and then switching to $\mathbb{F}$ via Figure 2, we observe that the within-word part of the claim is a public vector $\omega \in \mathbb{T}^{2^\iota}$ of weights on the 2^ι partial evaluations t , and run the switch on the bit-level claim directly, with $\omega^\top t$ in place of $\theta^\top t$ in the verifier’s check. This saves one sumcheck of ν rounds and one message of 2^ι elements of $\mathbb{T}$. Thus the two integrations share the commitment and opening implementation while retaining their respective outer protocols. The two statements differ in shape. For Binius the circuit is a single hash of one long message, so its compressions form one chain and each consumes the state the previous one produced. For Flock the statement is a batch of independent compression calls with no chaining between them. We could not integrate BinarySpartan [Set26], as its code was not available at the time of writing.

We remark that the codebases of Binius and Flock differ significantly from those described in the publications [DP25, BRW26] (e.g. Binius got enriched with some ideas from Flock, including *univariate skip*, and Flock does not support Keccak-256 proofs). They are under active development⁴. We made a few modifications to the Flock prover to ensure that our comparison is fair. First, Flock uses Ligerito in a default profile targeting about 128 bits of security. We modify it to aim for 100 bits of security at rate 1/2. Second Flock defaults to BLAKE3 for its Merkle trees, which is not optimal on our machine. We therefore use hardware-accelerated SHA-256 Merkle trees for Flock and Ligerito throughout, which is the faster choice for them on the hardware used in our experiments.

5.1.6 Comparison with Other Polynomial Commitments

We compare our implementation with the BaseFold [ZCF24], WHIR [ACFY25], and Ligerito [NA25] polynomial commitments. All of those PCS are field-agnostic, and for fairness of comparison we instantiate them over the GHASH field. For BaseFold and WHIR, the implementation is taken from the Binius codebase [Irr26], and for Ligerito we use the implementation from the Flock repository [BRW26]. Binius has since removed its WHIR implementation. We keep WHIR in the comparison nonetheless, so the WHIR rows are built from a commit from the day before the removal, while every other row uses the commits stated in the citations. We use BaseFold, WHIR and Brakedown in the unique-decoding regime with no grinding at 100 bits, and Ligerito in the two Flock profiles built for 100 bits, which rely on the Johnson bound with out-of-domain samples and, at rate 1/4, on 16 bits of grinding per level. All three commit with Reed–Solomon codes, whose relative distance at rate ρ is $1 - \rho$, so the rate is a free parameter, and we commit at $\rho = 1/2$ and $\rho = 1/4$. A higher rate implies more queries, thus a larger opening proof, but faster commitment and opening time for a prover. We also compare against the tensor commitment of Brakedown [GLS⁺23], whose expander code encodes in a linear number of field operations rather than

⁴We use Binius at commit `81cdfbf` of 14 September 2026 and Flock at commit `b684b12` of 11 September 2026, the most recent at the time of writing, except for the WHIR rows of Table 1, where the PCS is taken from Binius at commit `37e9cd6`.

the quasilinear encoding of Reed–Solomon. No implementation of that code targets a binary field, so we implement it over GHASH field following the construction and the parameters of [GLS⁺23].

5.1.7 Recursion

The recursive mode links the C implementation of LaBRADOR [BS23] as a static library through a foreign-function interface. The prover and verifier construct the encoding of Appendix A.2. The recursive statement is bound to the preceding transcript by a digest of its absorbed data.

5.1.8 SIS Parameter Selection

We estimate the conventional lattice attack on each SIS instance with the Lattice Estimator [APS15] and target $\lambda = 100$ bits of security. In the *correctness* direction we use average-case estimates for the norms of the folded witness. For the *soundness* direction, we use the worst-case bound on the extracted preimage as stated in Section 3. For bit-dropped commitments, the length bound includes both the extracted folded witness and the reconstructed rounding residual through the combined bound $\sqrt{B_v^2 + B_z^2}$.

We account for the module-BKZ predictions of [DEdP25] and allow the attack to use every relevant subfield. For conductors of the form $2^a 3^b$, the lowest estimated cost is obtained over $\mathbb{Q}(\zeta_3)$. The conversion from the coefficient length bound to the canonical embedding assumes that a short vector has a random direction there, under which the conversion follows from the trace-form Gram matrix. We translate the resulting block size to its module-BKZ equivalent and evaluate it with the same reduction-cost model as the conventional attack.

5.2 Experimental Setup and Results

Every number below is a single thread pinned to one core of an AMD EPYC 9R05 on an AWS `m8azn.metal-12x1` instance with 192 GiB of RAM.

5.3 Polynomial Commitment Benchmarks

Table 1 shows the concrete sizes and timings of our implementation, as well as the other polynomial commitment schemes, i.e. BaseFold, WHIR, Brakedown and Ligerito. Commitment and prover times scale linearly with the number of elements for all of them, and at these sizes the quasilinear encoding of BaseFold, WHIR and Ligerito is not distinguishable from the linear encoding of Brakedown. Our scheme in non-recursive mode behaves the same way. In recursive mode the prover and the commitment are dominated by the LaBRADOR prover and commitment, which grow as the square root of the witness size, so their share of the total falls as the witness grows and the linear commitment takes over. Concretely, at the largest size our non-recursive commitment and prover times combined are within 25% of Brakedown and BaseFold, and between 1.6 and 2.3 times faster than WHIR and Ligerito.

Verification times remain almost flat for BaseFold, WHIR and Ligerito, and their proof sizes grow logarithmically. The proof of our recursive mode is constant across the whole range and the smallest in the table at every size, about three times smaller than the best of the others at the largest size. The proof size of Brakedown and of our non-recursive mode, and the verification time of Brakedown and of our both modes, grow roughly as the square root of the witness size. At the smallest size our non-recursive verification and proof size is comparable to the other schemes, with rising difference at larger sizes. In terms of verification time, our scheme in the recursive mode is not competitive, and we

leave the integration with a verifier-efficient lattice-based scheme, e.g. [KLN⁺26], as future work.

Table 1: Polynomial commitment schemes with various code rates stated if applicable. Times are in milliseconds, sizes of commitment C and proof π are in KiB. All rows target 100 bits. Towards our disadvantage, our scheme is instantiated over $\mathbb{F}_{2^{162}}$ and the other schemes are instantiated over $\mathbb{F}_{2^{128}}$.

	$\mathbb{F}$ elements: 2^{18}					$\mathbb{F}$ elements: 2^{20}				
	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ C $	$ \pi $	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ C $	$ \pi $
BaseFold, 1/2	5.6	2.4	0.3	0.03	243.8	25.0	9.3	0.4	0.03	312.2
BaseFold, 1/4	11.1	3.4	0.2	0.03	184.3	49.3	13.1	0.3	0.03	235.3
WHIR, 1/2	7.1	8.5	2.2	0.03	190.2	31.3	48.0	1.6	0.03	239.7
WHIR, 1/4	11.3	5.6	0.9	0.03	154.4	61.5	30.1	0.8	0.03	202.8
Ligerito, 1/2	4.4	12.9	0.6	4.0	350.2	26.1	50.3	1.1	4.0	378.9
Ligerito, 1/4	9.0	16.7	0.4	2.0	176.5	53.4	55.9	0.7	2.0	191.7
Brakedown, 0.704	6.5	2.9	5.6	0.03	3498.6	23.6	9.1	10.9	0.03	6158.0
Brakedown, 0.581	6.7	2.0	3.2	0.03	1658.5	29.8	6.8	6.2	0.03	3003.9
.....										
LaBinius	7.0	2.8	0.3	283.5	307.5	42.7	10.8	1.2	587.2	658.1
LaBinius + LaBRADOR	9.8	121.7	76.8	4.1	75.3	48.1	300.9	192.8	4.1	72.6
.....										
	$\mathbb{F}$ elements: 2^{22}					$\mathbb{F}$ elements: 2^{24}				
	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ C $	$ \pi $	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ C $	$ \pi $
BaseFold, 1/2	110.5	40.9	0.5	0.03	395.1	468.7	175.4	0.6	0.03	478.5
BaseFold, 1/4	213.6	55.3	0.3	0.03	285.0	907.0	243.8	0.4	0.03	345.3
WHIR, 1/2	133.6	127.7	1.6	0.03	294.0	649.9	909.9	2.0	0.03	352.7
WHIR, 1/4	261.9	143.4	1.3	0.03	243.4	1069.4	594.5	1.1	0.03	300.9
Ligerito, 1/2	96.8	240.5	1.3	4.0	330.8	482.9	837.8	1.0	4.0	469.5
Ligerito, 1/4	199.5	284.4	0.8	2.0	183.9	963.5	917.9	0.6	2.0	249.2
Brakedown, 0.704	120.3	27.4	20.8	0.03	11149.9	604.7	97.0	40.2	0.03	20806.5
Brakedown, 0.581	155.3	23.3	13.5	0.03	5602.0	978.3	86.4	27.1	0.03	10705.6
.....										
LaBinius	164.7	44.8	2.7	1174.5	1394.9	636.0	167.3	6.7	2430.0	2927.3
LaBinius + LaBRADOR	174.2	603.4	370.4	4.1	86.1	642.9	1333.8	713.1	4.5	82.9

5.4 Binius and Flock as the outer proof systems

We use the codebase of Binius [Irr26] and Flock [BRW26] in semi black-box manner, i.e. we separate its *information-theoretic* reductions used to turn the original statement into an evaluation claim over $\mathbb{T}$ from its *cryptographic* commitment and opening. More precisely, we use its reductions and the witness generation, but replace its Merkle-tree commitment over a Reed–Solomon codeword and BaseFold or Ligerito opening with our Ajtai commitment and PCS from Section 3. The resulting evaluation claim over $\mathbb{T}$ is passed through the field switch (Figure 2) to $\mathbb{F}$ for our commitment to open.

Tables 2 to 4 report Keccak-256, SHA-256 and BLAKE3 under Binius, and Tables 5 and 6 report BLAKE3 and SHA-256 under Flock. In those benchmarks, we do not distinguish prover and commitment times as we envision that in practical applications we mainly care about general throughput of the proof generation. For Binius, the prover times exclude witness generation, and for Flock we include generation as in the original codebases. For Flock we run its single-table path, which commits the full padded witness, since that is the path in which our integration replaces the commitment.

For Binius, at the largest parameters we consider and for all three hashes, i.e. Keccak-256, SHA-256 and BLAKE3, our non-recursive prover costs at most 1.2 times the original one and our non-recursive verifier less than 1.1 times, our recursive prover less than 1.5

times and our recursive verifier less than 1.4 times. At those parameters our recursive proof is about five times shorter than the Binius proof. The Binius verifier evaluates the constraint system directly, which walks every constraint and is therefore not succinct, as stated in [Irr26]. Without it, verification takes between 78 and 132 ms for Binius, between 89 and 142 ms for our non-recursive mode and between 793 and 835 ms for our recursive mode.

For Flock, at the largest parameters and for both hashes, our non-recursive prover takes about 0.9 of the time of the original one and our non-recursive verifier about 1.4 times, while our recursive prover costs less than 1.5 times the original. Our recursive proof is about five times shorter than the Flock proof. The recursive verifier is the exception, at 26 to 27 times the Flock verifier, because the Flock verifier stays at a few tens of milliseconds at every size while ours grows as square-root with the witness size. Overall, the Flock arithmetisation seems to offer more throughput for the prover. In the most favourable benchmark, we process 4.3 MB of data per second for BLAKE3 with Flock in non-recursive mode and 2.7 MB per second in the recursive mode.

Table 2: Keccak-256 under Binius; prover times exclude witness generation. Times are in milliseconds, sizes are in KiB.

	873 permutations			⋮	3 495 permutations		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $		$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Binius	165.0	95.6	244.1	⋮	667.2	390.6	310.4
.....							
LaBinius	167.3	96.2	584.8	⋮	716.3	394.6	1286.7
LaBinius + LaBRADOR	275.7	168.9	89.1	⋮	1011.1	595.2	86.3
	13 981 permutations			⋮	55 924 permutations		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $		$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Binius	2671.0	1564.7	390.5	⋮	10821.3	6250.7	471.3
.....							
LaBinius	3187.2	1580.4	2564.1	⋮	12166.0	6266.6	5452.5
LaBinius + LaBRADOR	3641.7	1948.3	100.1	⋮	13828.4	6969.7	97.6

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Table 3: SHA-256 under Binius; prover times exclude witness generation. Times are in milliseconds, sizes are in KiB.

	1 134 compressions			4 539 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Binius	72.3	36.4	244.2	297.2	144.9	310.5
LaBinius	75.0	37.0	591.4	356.2	153.2	1273.4
LaBinius + LaBRADOR	196.2	112.2	88.9	654.3	359.1	86.7
	18 157 compressions			72 628 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Binius	1214.2	580.2	390.6	4853.6	2315.2	471.4
LaBinius	1480.5	632.7	2510.7	5753.1	2488.6	5380.1
LaBinius + LaBRADOR	2069.1	999.4	100.3	6923.1	3191.2	97.7

Table 4: BLAKE3 under Binius; prover times exclude witness generation. Times are in milliseconds, sizes are in KiB.

	2 055 compressions			8 225 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Binius	99.0	50.4	244.2	401.4	201.5	310.5
LaBinius	100.6	51.0	607.3	434.6	209.6	1261.2
LaBinius + LaBRADOR	210.9	127.6	88.7	685.6	400.1	86.6
	32 906 compressions			131 624 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Binius	1628.3	809.7	390.6	6544.8	3240.4	471.4
LaBinius	1870.0	862.0	2633.5	7334.4	3399.9	5417.8
LaBinius + LaBRADOR	2441.8	1223.2	100.2	8485.8	4091.2	97.6

Table 5: BLAKE3 under Flock; prover times include witness generation, which Flock performs inside its prover. Times are in milliseconds, sizes are in KiB.

	2 048 compressions			8 192 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Flock	55.3	26.2	369.8	148.2	27.2	398.6
LaBinius	58.1	27.6	565.2	147.4	28.9	1196.3
LaBinius + LaBRADOR	175.8	112.3	88.5	439.5	223.6	85.6
	32 768 compressions			131 072 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Flock	560.8	27.9	351.0	2121.8	27.3	490.0
LaBinius	505.8	31.7	2450.7	1953.9	38.6	5130.3
LaBinius + LaBRADOR	1075.3	396.4	99.2	3072.4	729.0	96.4

Table 6: SHA-256 under Flock; prover times include witness generation, which Flock performs inside its prover. Times are in milliseconds, sizes are in KiB.

	1 024 compressions			4 096 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Flock	56.4	27.2	369.9	149.5	28.5	398.7
LaBinius	54.0	27.4	542.0	148.5	29.9	1146.2
LaBinius + LaBRADOR	181.7	111.5	88.2	442.8	225.2	85.7
	16 384 compressions			65 536 compressions		
	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Flock	575.1	28.2	351.0	2184.1	27.8	490.1
LaBinius	527.2	32.5	2322.3	2036.8	38.7	4848.6
LaBinius + LaBRADOR	1093.5	392.9	99.3	3172.3	730.7	96.5

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A Details of the LaBRADOR Recursion

The final message of Π^{fold} reveals the folded witness $\mathbf{v} \in \mathcal{R}^m$, the commitment matrix $\mathbf{Y}$, and the left expansion $\mathbf{u}$. They satisfy

$$\mathbf{F}\mathbf{v} = \mathbf{Y}\mathbf{c} \bmod q, \quad \mathbf{B}\mathbf{v} = \mathbf{u}^\top \mathbf{c} \bmod 2, \quad \|\mathbf{v}\| \leq \beta'.$$

We replace this message by commitments followed by a proof in LaBRADOR. The matrix $\mathbf{Y}$ is committed before the folding challenge through the residue number system representation of Definition 34. Each residue matrix occupies one witness vector. The left expansion $\mathbf{u}$ is committed before the folding challenge. The recursive proof opens these commitments and establishes the folded relation through the system of constraints.

A.1 The LaBRADOR Relation

We recall the relation proved by LaBRADOR [BS23]. Let Q be a prime and let $f' = 2d$ be a power of two. Write

$$\mathcal{R}'_Q := \mathbb{Z}[\zeta_{f'}]/Q\mathbb{Z}[\zeta_{f'}] \cong \mathbb{Z}_Q[x] / (x^d + 1).$$

Definition 25 (LaBRADOR Relation). Let $r, K, L \in \mathbb{N}$. An instance is

$$\mathcal{I}_{\text{Lab}} := \left(\left\{ a_{i,j}^{(k)}, \phi_i^{(k)}, b^{(k)} \right\}_{k \in [K]}, \left\{ f'^{(\ell)} \right\}_{\ell \in [L]}, \mathcal{I}_{\text{bin}}, \{\beta_i\}_{i \in [r]} \right),$$

where $a_{i,j}^{(k)}, b^{(k)} \in \mathcal{R}'_Q$, $\phi_i^{(k)} \in \mathcal{R}'_Q^{n_i}$, every $f'^{(\ell)}$ has the same form as $f^{(k)}$, and $\mathcal{I}_{\text{bin}} \subseteq [r]$ designates the binary witness vectors. Its witness is

$$\mathbf{s} = (\mathbf{s}_0, \dots, \mathbf{s}_{r-1}), \quad \mathbf{s}_i \in \mathcal{R}'_Q^{n_i}.$$

Define

$$f^{(k)}(\mathbf{s}) := \sum_{i,j \in [r]} a_{i,j}^{(k)} \langle \mathbf{s}_i, \mathbf{s}_j \rangle + \sum_{i \in [r]} \langle \phi_i^{(k)}, \mathbf{s}_i \rangle - b^{(k)}.$$

The relation is

$$\mathcal{L}_{\text{LaBRADOR}} := \left\{ (\mathcal{I}_{\text{Lab}}, \mathbf{s}) : \begin{array}{ll} f^{(k)}(\mathbf{s}) = 0 \in \mathcal{R}'_Q & \text{for every } k \in [K], \\ \text{ct}(f^{(\ell)}(\mathbf{s})) = 0 \in \mathbb{Z}_Q & \text{for every } \ell \in [L], \\ \text{coeff}(\mathbf{s}_i) \in \{0, 1\}^{n_i} & \text{for every } i \in \mathcal{I}_{\text{bin}}, \\ \|\mathbf{s}_i\|^2 \leq \beta_i^2 & \text{for every } i \in [r] \end{array} \right\}^5.$$

A.2 Encoding Cyclotomic Arithmetic

We specialise the relations from Section 3 to the cyclotomic ring $\mathcal{R} = \mathbb{Z}[\zeta_f]$ of conductor $f = 3^\kappa$, where $\kappa \in \mathbb{N}$, so that $\varphi = 2 \cdot 3^{\kappa-1}$ and

$$\mathcal{R}_Q \cong \mathbb{Z}_Q[x] / (\Phi_f(x)), \quad \Phi_f(x) = x^\varphi + x^{\varphi/2} + 1.$$

The folded constraints use the moduli q and 2. We first express them as integer identities in $\mathcal{R}$ and then reduce them modulo the separate LaBRADOR modulus Q , which gives identities in $\mathcal{R}_Q$. The two cyclotomic rings modulo Q do not match. Elements of $\mathcal{R}_Q$ have φ coefficients, while elements of $\mathcal{R}'_Q$ have d coefficients, where d is a power of two and smaller than φ . An element of $\mathcal{R}_Q$ does not fit into one LaBRADOR witness element, and a product in $\mathcal{R}_Q$ is not a product in $\mathcal{R}'_Q$.

We work with coefficients only. A witness element $x \in \mathcal{R}_Q$ is cut into chunks of c consecutive coefficients, and every chunk is stored as one element of $\mathcal{R}'_Q$. For a public element $g \in \mathcal{R}_Q$, the product gx is, before reduction modulo Φ_f , the sum over all chunks of the product of the chunk with a shifted copy of g . To keep every such product inside $\mathcal{R}'_Q$, the shifted copies of g are cut as well, into blocks of p consecutive coefficients. The product of a block with a chunk has at most $p + c - 1 \leq d$ coefficients, so LaBRADOR computes it exactly, without reduction modulo $x^d + 1$.

An identity $\sum_\nu g_\nu x_\nu = z$ in $\mathcal{R}_Q$ then becomes one equation per block of p coefficient positions, in the same way that two long integers are added digit by digit. There are $N := \varphi/p$ blocks, indexed by $a \in [N]$, and block a covers positions $ap, \dots, ap + p - 1$. The equation for block a collects the block products that start at position ap . Such a product has $p + c - 1$ coefficients, so it also covers the first $c - 1$ positions of block $a + 1$. This excess is moved into the equation of block $a + 1$ through a carry e_a : the equation of block a subtracts $x^p e_a$, and the equation of block $a + 1$ adds e_a . The carry out of the last block covers positions φ and above, which do not exist in $\mathcal{R}_Q$. Reduction modulo Φ_f replaces x^φ by $-x^{\varphi/2} - 1$, so this last carry is instead subtracted once in block 0 and once in block $N/2$. Figure 5 shows the layout.

Definition 26 (Coefficient Representatives and Support). Every element of $\mathcal{R}'_Q$ is identified with its unique representative in $\mathbb{Z}_Q[x]$ of degree below d , and every element of $\mathcal{R}_Q$ with its unique representative in $\mathbb{Z}_Q[x]$ of degree below φ . For an element $y = \sum_t y_t x^t$ of either ring, y_t is its t -th coefficient, with $y_t := 0$ beyond the degree of the representative, and its support is

$$\text{supp}(y) := \{t : y_t \neq 0\}.$$

⁵The binary witness-vector set $\mathcal{I}_{\text{bin}}$ are formally conventions of the Dachshund front end of the LaZer library [LSS24], whose LaBRADOR implementation our recursive mode links (Section 5).

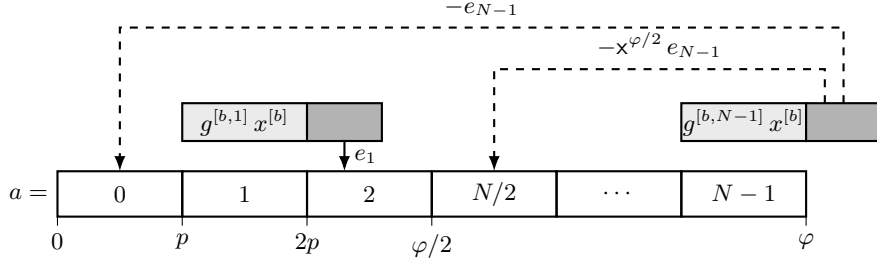


Figure 5: Coefficient blocks, block products, and carries of the block constraint system.

An identity between elements of $\mathcal{R}'_Q$ or of $\mathcal{R}_Q$ holds in $\mathbb{Z}_Q[x]$ if it holds between their representatives.

Definition 27 (Block Parameters). The chunk length c and the block length p are positive integers with

$$c + p - 1 \leq d, \quad p \mid \frac{\varphi}{2}.$$

The number of blocks is $N := \varphi/p$, the block index set is $\mathcal{A} := [N]$, and the chunk index set is $\mathcal{B} := \lceil \varphi/c \rceil$.

Definition 28 (Coefficient Blocks). For $y \in \mathcal{R}_Q$ and $a \in \mathcal{A}$, the a -th coefficient block of y is

$$y^{(a)} := \sum_{t \in [p]} y_{ap+t} x^t \in \mathcal{R}'_Q.$$

Every coefficient block has support in $[p]$, and since $Np = \varphi$,

$$y = \sum_{a \in \mathcal{A}} x^{ap} y^{(a)} \in \mathbb{Z}_Q[x]. \quad (2)$$

Definition 29 (Witness Chunks). For $x \in \mathcal{R}_Q$ and $b \in \mathcal{B}$, the b -th witness chunk of x is

$$x^{[b]} := \sum_{t \in [c]} x_{bc+t} x^t \in \mathcal{R}'_Q.$$

Definition 30 (Chunk Encoding and Decoding). The chunk encoding map $\text{Enc}_c: \mathcal{R}_Q \rightarrow \mathcal{R}'_Q^{|\mathcal{B}|}$ and the chunk decoding map $\text{Dec}_c: \mathcal{R}'_Q^{|\mathcal{B}|} \rightarrow \mathcal{R}_Q$ are

$$\text{Enc}_c(x) := (x^{[b]})_{b \in \mathcal{B}}, \quad \text{Dec}_c((\xi^{[b]})_{b \in \mathcal{B}}) := \sum_{b \in \mathcal{B}} x^{bc} \xi^{[b]} \bmod \Phi_f(x).$$

Both maps are applied entrywise to vectors and matrices.

Every witness chunk has support in $[c]$, and $\text{Dec}_c(\text{Enc}_c(x)) = x$ for every $x \in \mathcal{R}_Q$, because $\sum_{b \in \mathcal{B}} x^{bc} x^{[b]} = \sum_{u \in [\varphi]} x_u x^u$ in $\mathbb{Z}_Q[x]$. The map Dec_c is also defined on chunk families whose members have coefficients outside $[c]$, and the block equations below are sound only for families that do not.

Definition 31 (Public Coefficient Blocks). For a public $g \in \mathcal{R}_Q$, $b \in \mathcal{B}$, and $a \in \mathcal{A}$, the public coefficient block of g at (b, a) is

$$g^{[b,a]} := (x^{bc} g)^{(a)} \in \mathcal{R}'_Q,$$

the a -th coefficient block of the ring element $x^{bc} g \in \mathcal{R}_Q$.

Lemma 12 (Chunk Products). *Let $g \in \mathcal{R}_Q$, let $b \in \mathcal{B}$, and let $\xi \in \mathcal{R}'_Q$ have support in $[c]$. For every $a \in \mathcal{A}$, the product $g^{[b,a]} \xi$ in $\mathcal{R}'_Q$ equals the product of the representatives in $\mathbb{Z}_Q[x]$ and has support in $[p + c - 1]$. Moreover,*

$$\sum_{a \in \mathcal{A}} x^{ap} g^{[b,a]} \xi = x^{bc} g \xi \in \mathbb{Z}_Q[x].$$

Proof. The representatives of $g^{[b,a]}$ and ξ have degrees below p and below c , so their product in $\mathbb{Z}_Q[x]$ has degree at most $p + c - 2 \leq d - 1$ by Definition 27. This product is its own representative modulo $x^d + 1$, which is the first claim. For the second claim, Equation (2) applied to $x^{bc} g \in \mathcal{R}_Q$ gives $\sum_{a \in \mathcal{A}} x^{ap} g^{[b,a]} = x^{bc} g$ in $\mathbb{Z}_Q[x]$, and it remains to multiply by ξ . $\square$

Definition 32 (Block Constraint System). Let $m \in \mathbb{N}$, let $g_\nu \in \mathcal{R}_Q$ for $\nu \in [m]$ and $z \in \mathcal{R}_Q$ be public, and consider the identity

$$\sum_{\nu \in [m]} g_\nu x_\nu = z \in \mathcal{R}_Q$$

in the witnesses $x_\nu \in \mathcal{R}_Q$. Its block constraint system has as witness the chunks $x_\nu^{[b]} \in \mathcal{R}'_Q$ for $\nu \in [m]$ and $b \in \mathcal{B}$ together with the carries $e_a \in \mathcal{R}'_Q$ for $a \in \mathcal{A}$, where e_a is the carry out of block a and $e_{-1} := 0$. With the block product sums

$$D_a := \sum_{\nu \in [m]} \sum_{b \in \mathcal{B}} g_\nu^{[b,a]} x_\nu^{[b]} \in \mathcal{R}'_Q, \quad a \in \mathcal{A},$$

the system consists of the equations

$$D_a + e_{a-1} - x^p e_a - \mathbf{1}_{\{a=0\}} e_{N-1} - \mathbf{1}_{\{a=N/2\}} e_{N-1} = z^{(a)} \in \mathcal{R}'_Q \quad (\text{block})$$

for every $a \in \mathcal{A}$.

Its support conditions are $\text{supp}(x_\nu^{[b]}) \subseteq [c]$ for every $\nu \in [m]$ and $b \in \mathcal{B}$, and $\text{supp}(e_a) \subseteq [d-p]$ for every $a \in \mathcal{A}$.

Every equation in Equation (block) is linear over $\mathcal{R}'_Q$ in the witness, because one factor of every product in D_a is public.

Lemma 13 (Block Encoding). *Let the chunks $x_\nu^{[b]}$ satisfy the support conditions of Definition 32. There exist carries $e_a \in \mathcal{R}'_Q$ with $\text{supp}(e_a) \subseteq [d-p]$ for every $a \in \mathcal{A}$ that satisfy Equation (block) if and only if*

$$\sum_{\nu \in [m]} g_\nu \text{Dec}_c((x_\nu^{[b]})_{b \in \mathcal{B}}) = z \in \mathcal{R}_Q.$$

Proof. Let

$$P := \sum_{a \in \mathcal{A}} x^{ap} D_a \in \mathbb{Z}_Q[x].$$

By Lemma 12, every product $g_\nu^{[b,a]} x_\nu^{[b]}$ is a product in $\mathbb{Z}_Q[x]$ of degree at most $p + c - 2$, so D_a has degree at most $p + c - 2$ and P has degree at most $p + c - 2$. Summing the second claim of Lemma 12 over $b \in \mathcal{B}$ and $\nu \in [m]$ gives

$$P = \sum_{\nu \in [m]} g_\nu \sum_{b \in \mathcal{B}} x^{bc} x_\nu^{[b]} \in \mathbb{Z}_Q[x],$$

so by Definition 30 the identity in the statement holds if and only if $P = z \bmod \Phi_f(x)$.

Suppose first that carries with the stated support satisfy Equation (block). In the equation for block a , the terms D_a , e_{a-1} , $x^p e_a$, e_{N-1} , and $z^{(a)}$ have degrees at most $p + c - 2$, $d - p - 1$, $d - 1$, $d - p - 1$, and $p - 1$, all below d , so the equation holds in $\mathbb{Z}_Q[x]$. Multiply the equation for block a by x^{ap} and sum over $a \in \mathcal{A}$. Since $e_{-1} = 0$, the carry terms contribute

$$\sum_{a \in \mathcal{A}} x^{ap} e_{a-1} - \sum_{a \in \mathcal{A}} x^{(a+1)p} e_a = \sum_{a \in [N-1]} x^{(a+1)p} e_a - \sum_{a \in [N]} x^{(a+1)p} e_a = -x^\varphi e_{N-1},$$

and the two indicator terms contribute $-e_{N-1} - x^{\varphi/2} e_{N-1}$. Together with Equation (2) for z , the sum reads

$$P - \Phi_f(x) e_{N-1} = z \in \mathbb{Z}_Q[x],$$

so $P = z \bmod \Phi_f(x)$.

Conversely, suppose that $P = z \bmod \Phi_f(x)$. Since Φ_f is monic of degree φ , z has degree below φ , and P has degree at most $\varphi + c - 2$, there is a unique $W \in \mathbb{Z}_Q[x]$ with

$$P - z = \Phi_f(x) W, \quad \deg W \leq c - 2.$$

Set

$$D'_a := D_a - \mathbf{1}_{\{a=0\}} W - \mathbf{1}_{\{a=N/2\}} W \quad \text{for } a \in \mathcal{A},$$

so that $\sum_{a \in \mathcal{A}} x^{ap} D'_a = P - W - x^{\varphi/2} W = z + x^\varphi W$. For $a \in \mathcal{A}$ define

$$e_a := x^{-(a+1)p} \left(\sum_{a' \in [a+1]} x^{a'p} D'_{a'} - \sum_{a' \in [a+1]} x^{a'p} z^{(a')} \right).$$

The term $x^{a'p} D'_{a'}$ has support in positions at least $a'p$, and $x^\varphi W$ has support in positions at least $\varphi = Np$. Hence the coefficients of $\sum_{a' \in [a+1]} x^{a'p} D'_{a'}$ at positions below $(a+1)p$ agree with those of $z + x^\varphi W$, which are the coefficients of z , which in turn agree with those of $\sum_{a' \in [a+1]} x^{a'p} z^{(a')}$ by Equation (2). The bracket therefore vanishes at all positions below $(a+1)p$, and e_a is an element of $\mathbb{Z}_Q[x]$. The bracket has degree at most $ap + p + c - 2$, so $\deg e_a \leq c - 2 \leq d - p - 1$, that is, $\text{supp}(e_a) \subseteq [d - p]$. For $a = N - 1$ the bracket equals $z + x^\varphi W - z$, so $e_{N-1} = W$. Finally, subtracting the defining equations of e_a and e_{a-1} , and using $e_{-1} = 0$ for $a = 0$,

$$x^{(a+1)p} e_a - x^{ap} e_{a-1} = x^{ap} (D'_a - z^{(a)}) \quad \text{for every } a \in \mathcal{A},$$

and dividing by x^{ap} gives $D'_a + e_{a-1} - x^p e_a = z^{(a)}$ in $\mathbb{Z}_Q[x]$. With $e_{N-1} = W$ this is the equation for block a in Equation (block), and it holds in $\mathcal{R}'_Q$ after reduction modulo $x^d + 1$. $\square$

The support conditions are necessary. A chunk with a nonzero coefficient at a position in $[c, d)$, or a carry with one at a position in $[d - p, d)$, turns the equation for its block into an identity modulo $x^d + 1$ only, and the sum in the proof is then no longer an identity in $\mathbb{Z}_Q[x]$.

A.3 Zero Parts and Integer Identities

The high coefficients of every encoded witness chunk must be zero. Let $\mathcal{Z}$ be the set of pairs (ξ, j) consisting of a witness chunk ξ and one forbidden coefficient position j . After all chunk commitments are fixed, the transcript derives independent uniform scalars

$$\rho_{\xi,j}^{(h)} \in \mathbb{Z}_Q, \quad h \in \left[\left\lceil \frac{\lambda}{\log_2 Q} \right\rceil \right].$$

For every such h , the statement contains the constant-term constraint

$$\text{ct} \left(\sum_{(\xi,j) \in \mathcal{Z}} \rho_{\xi,j}^{(h)} x^{-j} \xi \right) = 0 \in \mathbb{Z}_Q. \quad (\text{zero})$$

Since $\text{ct}(x^{-j} \xi)$ is the j -th coefficient of ξ , Equation (zero) is a random linear combination of all forbidden coefficients.

Lemma 14 (Soundness of the Zero-Part Test). *Assume that the Ajtai commitments binding the chunks are binding before the scalars in Equation (zero) are derived. If at least one forbidden coefficient is nonzero, all zero constraints hold with probability at most $2^{-\lambda}$.*

Proof. Binding fixes every forbidden coefficient before the scalars are sampled. For one repetition, the left-hand side of Equation (zero) is a nonzero linear form in independent uniform elements of the field $\mathbb{Z}_Q$. It is zero with probability Q^{-1} . The repetitions are independent, so the probability that all constraints hold is

$$Q^{-\lceil \lambda / \log_2 Q \rceil} \leq 2^{-\lambda}.$$

□

Definition 33 (Base- B Gadget Decomposition). Let $B \geq 2$, and define the balanced digit set

$$\mathcal{D}_B := \left\{ -\left\lfloor \frac{B}{2} \right\rfloor, \dots, \left\lceil \frac{B}{2} \right\rceil - 1 \right\}.$$

A base- B gadget decomposition of an object x with n_x coefficients is a family

$$(x^{(j)})_{j \in [J_x]}$$

such that

$$x = \sum_{j \in [J_x]} B^j x^{(j)},$$

and every coefficient of every $x^{(j)}$ belongs to $\mathcal{D}_B$. For a vector or a matrix, this equality applies coefficientwise. Each digit object is represented by a separate witness vector. If $\mathbf{x}^{(j)}$ is its flattened coefficient vector, its norm cap is

$$\|\mathbf{x}^{(j)}\| \leq \frac{B-1}{2} \sqrt{n_x}.$$

A.4 The Encoded Folded Relation

Let

$$q = \prod_{\ell \in [\ell_{\max}]} q_\ell$$

be a product of distinct odd primes.

Definition 34 (RNS Representation). For $x \in \mathcal{R}_q$, its residue number system representation is the family

$$(x_\ell)_{\ell \in [\ell_{\max}]}$$

defined by the Chinese remainder isomorphism

$$\mathcal{R}_q \cong \prod_{\ell \in [\ell_{\max}]} \mathcal{R}_{q_\ell}, \quad x \mapsto (x_\ell)_{\ell \in [\ell_{\max}]}.$$

For every $\ell \in [\ell_{\max}]$, we identify x_ℓ with the unique lift to $\mathcal{R}$ whose canonical coefficients lie in

$$\left[-\frac{q_\ell - 1}{2}, \frac{q_\ell - 1}{2} \right].$$

For a vector or a matrix, the notation $(\cdot)_\ell$ is applied entrywise.

Let $\mathbf{F}_\ell$ and $\mathbf{Y}_\ell$ be the balanced residue representations of $\mathbf{F}$ and $\mathbf{Y}$. The limb constraint family is

$$\mathbf{F}_\ell \mathbf{v} - \mathbf{Y}_\ell \mathbf{c} = q_\ell \mathbf{k}_\ell \in \mathcal{R} \quad \text{for every } \ell \in [\ell_{\max}]. \quad (\text{limb})$$

Lemma 15 (RNS Lifting). *The folded identity*

$$\mathbf{F} \mathbf{v} = \mathbf{Y} \mathbf{c} \bmod q$$

holds if and only if there exist vectors $\mathbf{k}_\ell \in \mathcal{R}^n$ satisfying Equation (limb) for every $\ell \in [\ell_{\max}]$.

Proof. Suppose first that

$$\mathbf{F} \mathbf{v} = \mathbf{Y} \mathbf{c} \bmod q.$$

Reducing this identity modulo q_ℓ gives

$$\mathbf{F}_\ell \mathbf{v} = \mathbf{Y}_\ell \mathbf{c} \bmod q_\ell.$$

Every coefficient of

$$\mathbf{F}_\ell \mathbf{v} - \mathbf{Y}_\ell \mathbf{c}$$

is therefore divisible by q_ℓ . Dividing its coefficients by q_ℓ defines $\mathbf{k}_\ell \in \mathcal{R}$ satisfying Equation (limb).

Conversely, reducing every identity in Equation (limb) modulo q_ℓ gives

$$\mathbf{F}_\ell \mathbf{v} = \mathbf{Y}_\ell \mathbf{c} \bmod q_\ell.$$

The Chinese remainder isomorphism of Definition 34 combines these identities into

$$\mathbf{F} \mathbf{v} = \mathbf{Y} \mathbf{c} \bmod q.$$

□

Let

$$\tilde{\mathbf{u}} := \text{lift}(\mathbf{u}).$$

Let $\mathbf{e}_j := \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_1)_j$ for $j \in [r]$ be the public coefficients of the evaluation check $\mathbf{u}^\top \tilde{\mathbf{e}}\mathbf{q}(\mathbf{r}_1) = t$, and let t be the claimed evaluation. The binary constraint family is

$$\begin{aligned} \text{lift}(\mathbf{B}) \mathbf{v} - \sum_{j \in [r]} c_j \tilde{\mathbf{u}}_j &= 2 \mathbf{w}_b, \\ \sum_{j \in [r]} \text{lift}(\mathbf{e}_j) \tilde{\mathbf{u}}_j - \text{lift}(t) &= 2 \mathbf{w}_e. \end{aligned} \quad (\text{bin})$$

For every $\ell \in [\ell_{\max}]$, define

$$\mathbf{y}_\ell := \text{Enc}_c(\mathbf{Y}_\ell).$$

Let n_Y be the number of coefficients of $\mathbf{Y}_\ell$. The balanced RNS representation gives the direct norm cap

$$\|\mathbf{y}_\ell\| \leq \beta_{Y,\ell} := \frac{q_\ell - 1}{2} \sqrt{n_Y} \quad \text{for every } \ell \in [\ell_{\max}]. \quad (\text{norm})$$

Thus the residue vectors are used directly as LaBRADOR witness vectors. They are not gadget decomposed.

The carries, limb quotients, and binary quotients are gadget decomposed. Let

$$\mathcal{I}_{\text{blk}} := \{\text{limb}_\ell : \ell \in [\ell_{\max}]\} \cup \{\text{bin}, \text{eval}\}.$$

For every $\delta \in \mathcal{I}_{\text{blk}}$, let $\mathbf{e}_{\delta,a}$ denote the vector of carries out of block a for the corresponding identity. We use

$$\begin{aligned} \mathbf{k}_\ell &= \sum_{j \in [J_{k,\ell}]} B^j \mathbf{k}_\ell^{(j)}, \\ \mathbf{w}_\tau &= \sum_{j \in [J_{w,\tau}]} B^j \mathbf{w}_\tau^{(j)} \quad \text{for every } \tau \in \{\mathbf{b}, \mathbf{e}\}, \\ \mathbf{e}_{\delta,a} &= \sum_{j \in [J_{e,\delta}]} B^j \mathbf{e}_{\delta,a}^{(j)} \quad \text{for every } \delta \in \mathcal{I}_{\text{blk}} \text{ and } a \in \mathcal{A}. \end{aligned}$$

The public level counts are chosen so that every honest quotient and carry admits the corresponding base- B gadget decomposition. Every digit vector receives the cap of Definition 33.

Let

$$\mathbf{y} := (\mathbf{y}_\ell)_{\ell \in [\ell_{\max}]}.$$

The pre-commitments are

$$\begin{aligned} T_Y &= \text{Com}_{\mathbf{h}_Y}(\mathbf{y}), \\ T_u &= \text{Com}_{\mathbf{h}_u}(\text{Enc}_c(\tilde{\mathbf{u}})), \\ T_R &= \text{Com}_{\mathbf{h}_R}(\mathbf{z}_R), \end{aligned} \tag{open}$$

where $\mathbf{z}_R$ contains the entrywise chunk encodings of $\mathbf{v}$, all gadget-digit vectors of the limb quotients, all gadget-digit vectors of the carries, and the gadget-digit vectors of $\mathbf{w}_\mathbf{b}$ and $\mathbf{w}_\mathbf{e}$.

Definition 35 (Encoded Folded Relation). The instance is

$$\begin{aligned} \mathcal{I}_{\text{rec}} &:= \left(\{\mathbf{F}_\ell\}_{\ell \in [\ell_{\max}]}, \mathbf{c}, \mathbf{B}, \{\text{lift}(\mathbf{e}_j)\}_{j \in [r]}, t, B, \right. \\ &\quad \left. T_Y, T_u, T_R, \left\{ \rho_{\xi,j}^{(h)} \right\}, \{\eta_i, \beta_i\}_{i \in [r]} \right). \end{aligned}$$

Its witness consists of the residue vectors

$$(\mathbf{y}_\ell)_{\ell \in [\ell_{\max}]},$$

the chunk encoding of $\mathbf{v}$, the chunk encoding of $\tilde{\mathbf{u}}$, and the separately capped gadget-digit vectors of every carry, every $\mathbf{k}_\ell$, $\mathbf{w}_\mathbf{b}$, and $\mathbf{w}_\mathbf{e}$. The vector encoding $\tilde{\mathbf{u}}$ belongs to the designated binary witness-vector set. The relation is

$$\mathcal{R}_{\text{rec}} := \left\{ (\mathcal{I}_{\text{rec}}, \mathbf{s}) : \begin{array}{l} \text{open holds for } T_Y, T_u, \text{ and } T_R, \\ \text{block encodes every identity in } \textit{Equations}(\text{limb}) \text{ and } (\text{bin}), \\ \text{bit holds for the witness vector encoding } \tilde{\mathbf{u}}, \\ \text{zero holds for every } h, \\ \|\mathbf{s}_i\|^2 = \eta_i \leq \beta_i^2 \quad \text{for every } i \in [r] \end{array} \right\}.$$

For every encoded linear constraint ν and output coefficient t , let $C_{\nu,t,\ell}$ be the public linear operator mapping $\mathbf{y}_\ell$ to its contribution to that coefficient. In a limb constraint, $C_{\nu,t,\ell}$ contains the coefficients of the folding challenge $\mathbf{c}$. Let $\mathcal{U}$ index every remaining non-gadget witness vector, and let $\mathbf{r}_{\nu,t,i}$ be the integer row multiplying $\mathbf{s}_i$ for $i \in \mathcal{U}$. For a gadget digit vector $\mathbf{x}^{(j)}$, let $\mathbf{r}_{\nu,t,x,j}$ be its integer row. Finally, let $b_{\nu,t}$ be the public constant coefficient.

Lemma 16 (No Wraparound). *Suppose that every encoded linear constraint satisfies*

$$\begin{aligned} & \sum_{\ell \in [\ell_{\max}]} \|C_{\nu,t,\ell}\|_{\text{op}} \frac{q\ell - 1}{2} \sqrt{n_Y} \\ & + \sum_{i \in \mathcal{U}} \|\mathbf{r}_{\nu,t,i}\| \beta_i \\ & + \sum_{(x,j) \in \mathcal{G}} \|\mathbf{r}_{\nu,t,x,j}\| \frac{B-1}{2} \sqrt{n_x} + |b_{\nu,t}| < \frac{Q}{2}, \end{aligned} \quad (\text{nowrap})$$

where $\mathcal{G}$ indexes all gadget-digit witness vectors. Every encoded linear constraint that holds in $\mathcal{R}'_Q$ then holds as an integer coefficient identity.

Proof. The integer value of coefficient t in constraint ν is

$$\begin{aligned} & \sum_{\ell \in [\ell_{\max}]} C_{\nu,t,\ell} \mathbf{y}_\ell + \sum_{i \in \mathcal{U}} \langle \mathbf{r}_{\nu,t,i}, \mathbf{s}_i \rangle \\ & + \sum_{(x,j) \in \mathcal{G}} \langle \mathbf{r}_{\nu,t,x,j}, \mathbf{x}^{(j)} \rangle - b_{\nu,t}. \end{aligned}$$

The operator-norm bound for the residue vectors, Cauchy–Schwarz for the remaining vectors, and the digit caps of Definition 33 bound its absolute value by the left-hand side of Equation (nowrap). This value is strictly below $Q/2$. If the coefficient is zero modulo Q , it is an integer multiple of Q in that interval, hence it is zero. $\square$

Lemma 17 (Reduction to LaBRADOR). *For every encoded folded instance $\mathcal{I}_{\text{rec}}$, there is a constraint system computable from $\mathcal{I}_{\text{rec}}$ for which Definition 35 is exactly the relation of Definition 25.*

Proof. Every open equality is a linear constraint with the corresponding public Ajtai key. Every block equation is linear because one factor of each product is public. The binary witness-vector constraint for $\tilde{\mathbf{u}}$ is a binary constraint of Definition 25. Every zero test is a constant-term constraint. The remaining norm equalities are the norm constraints of the LaBRADOR front end. Grouping every separately capped residue vector and gadget-digit vector into the witness vectors $\mathbf{s}_i$ gives exactly an instance of Definition 25. $\square$

Reduction modulo 2 is a ring homomorphism from $\mathcal{R}$ onto the field $\mathbb{F}$ of Lemma 1. The bit constraint gives

$$\tilde{\mathbf{u}} = \text{lift}(\mathbf{u}) \bmod 2.$$

Reducing Equation (bin) modulo 2 therefore gives

$$\mathbf{B} \mathbf{v} = \mathbf{u}^\top \mathbf{c} \bmod 2,$$

together with the evaluation check of the scheme. Since the folding challenges are ternary, $c_j \tilde{\mathbf{u}}_j$ reduces to

$$(c_j \bmod 2) \mathbf{u}_j.$$

A.5 Protocol and Security

The message $\mathbf{f}$ is committed as $\mathbf{Y} = \mathbf{FW} \bmod q$ with $\text{vec}(\mathbf{W}) = \text{lift}(\phi(\mathbf{f}))$. The commitment algorithm first fixes the RNS residue vectors and the pre-commitment T_Y . The evaluation protocol then proves an opening of T_Y and the requested evaluation claim.

The commitment phase is separate from the evaluation proof. The algorithm $\text{Com}(\text{pp}, \mathbf{f})$ returns T_Y and an opening state st containing the committed message, the Ajtai opening data with slack 1, the matrix $\mathbf{Y}$, and its RNS residue vectors. The interaction between P and V proves that this state opens T_Y and that the evaluation at $\mathbf{r}$ equals t . The extractor outputs an opening state satisfying $\text{Open}(\text{pp}, T_Y, \mathbf{f}, \text{st}) = 1$ for which the decoded binary evaluation constraint gives $\text{MLE}[\mathbf{f}](\mathbf{r}) = t$.

Theorem 2 (Security of the Recursive Opening). *Assume knowledge soundness of LaBRADOR [BS23], binding of the Ajtai pre-commitments under the SIS assumption used for the commitment scheme, and the hypotheses of Theorem 1. Suppose that Equation (nowrap) holds for every encoded linear constraint. Let κ_{Lab} be the knowledge error of LaBRADOR for the constraint system of Lemma 17. Then the protocol is complete with correctness error $\delta = (\nu + \iota)/|\mathbb{F}|$, and its evaluation proof is a proof of knowledge with knowledge error*

$$\kappa = \frac{2\nu + \iota}{|\mathbb{F}|} + \frac{r}{|\mathcal{C}|} + \kappa_{\text{Lab}} + 5 \cdot 2^{-\lambda}$$

against probabilistic polynomial-time provers for

$$\Xi^{\text{eval}} = \{(\text{pp}, (T_Y, \mathbf{r}, t), (\mathbf{f}, \text{st})) : \text{Open}(\text{pp}, T_Y, \mathbf{f}, \text{st}) = 1 \wedge \text{MLE}[\mathbf{f}](\mathbf{r}) = t\}.$$

Proof. For completeness, the folding reductions have correctness error $(\nu + \iota)/|\mathbb{F}|$ by Corollary 1, and the LaBRADOR proof adds none. An honest execution of $\text{Com}(\text{pp}, \mathbf{f})$ fixes the balanced residue matrices $(\mathbf{Y}_\ell)_{\ell \in [\ell_{\max}]}$ and their pre-commitment T_Y . The evaluation prover uses the exact limb quotients of Equation (limb), the exact binary quotients of Equation (bin), and carries satisfying the block constraints. The required carries exist by Lemma 13. The public level counts give gadget decompositions of every carry and quotient. The residue vectors satisfy Equation (norm), and every digit vector satisfies the cap of Definition 33. The vector $\tilde{\mathbf{u}} = \text{lift}(\mathbf{u})$ is binary, so it satisfies the bit constraint. The zero constraints hold identically, the open constraints hold by construction, and the announced squared norms equal the witness norms. The LaBRADOR verifier therefore accepts. The decoded Ajtai opening satisfies $\text{Open}(\text{pp}, T_Y, \mathbf{f}, \text{st}, 1) = 1$, and the binary evaluation constraint gives $\text{MLE}[\mathbf{f}](\mathbf{r}) = t$.

For knowledge soundness, the recursive proof replaces the final message of Π^{fold} , so it is a proof of knowledge for $\Xi_{\text{par}_{\text{out}}}^{\text{bin-lin}}$ whose knowledge error is κ_{Lab} for the LaBRADOR extractor, plus $2^{-\lambda}$ for the zero test, plus $2^{-\lambda}$ for each binding failure of T_Y , T_u , and T_R . Sequential composition with Corollary 1 adds $(2\nu + \iota)/|\mathbb{F}| + r/|\mathcal{C}|$, and the SIS branch of its extraction relation adds $2^{-\lambda}$ as in Theorem 1. The LaBRADOR extractor returns a witness satisfying Definition 35. Binding of T_Y and T_u fixes the committed RNS residue vectors and the binary left expansion before $\mathbf{c}$ is derived. Binding of T_R fixes every later chunk before the zero masks are derived. By Lemma 14, except with probability at most $2^{-\lambda}$, all extracted chunks have their claimed support. By Lemma 13, the extracted chunks decode to the limb and binary identities in Equations (limb) and (bin) over $\mathcal{R}_Q$. By Lemma 16, these identities hold over the integers.

Reducing Equation (limb) modulo q_ℓ for every $\ell \in [\ell_{\max}]$ and applying Lemma 15 gives

$$\mathbf{F} \mathbf{v} = \mathbf{Y} \mathbf{c} \bmod q.$$

The bit constraint gives

$$\tilde{\mathbf{u}} = \text{lift}(\mathbf{u}) \bmod 2.$$

Reducing Equation (bin) modulo 2 and using Definition 3 and lemma 1 yields the binary folded checks, including the claimed evaluation. Together with the norm bound and $\mathbf{F}\mathbf{v} = \mathbf{Y}\mathbf{c} \bmod q$, these checks are the conditions of **Open**, so they give an extracted state $(\mathbf{f}, \mathbf{st})$ satisfying

$$\text{Open}(\text{pp}, T_Y, \mathbf{f}, \mathbf{st}) = 1$$

and

$$\text{MLE}[\mathbf{f}](\mathbf{r}) = t.$$

Thus the extracted tuple belongs to Ξ^{eval} . □